Watch: Why we still think EUR/USD can reach 1.20

One of the biggest questions we get from clients is how we can be so constructive about the euro, despite what's going on with Iran, the US, and the subsequent oil and gas shocks. ING's Francesco Pesole explains our FX team's thinking.

Read more in our latest FX Talking here

Why we still think EUR/USD can reach 1.20

We still think, despite everything, that the euro can rise to 1.20 against the dollar this year. Our FX Strategist, Francesco Pesole, says our team believes the Fed will look through this inflationary 'bump' driven by rising oil and gas prices and will cut rates twice again this year. And that will have an amplified effect on EUR/USD for several reasons. And while oil prices are set to remain elevated, we don't see the same risks for gas in Europe. So, the euro may trade lower in the short term, but should recover fairly strongly.

Watch video

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THINK Ahead: Opening the rate-hike Pandora's box could unleash fresh turmoil

Central banks are flirting with rate hikes, and markets have gone crazy. Memories of the 2022 inflation battle are clearly still fresh. Yet talking about rate rises is one thing, delivering them is quite another, and they could win the 'wrong' inflation fight. The bar is higher than markets think, says James Smith, as we look ahead to next week

Pandora unleashed the world's evils. We might be stretching this a bit

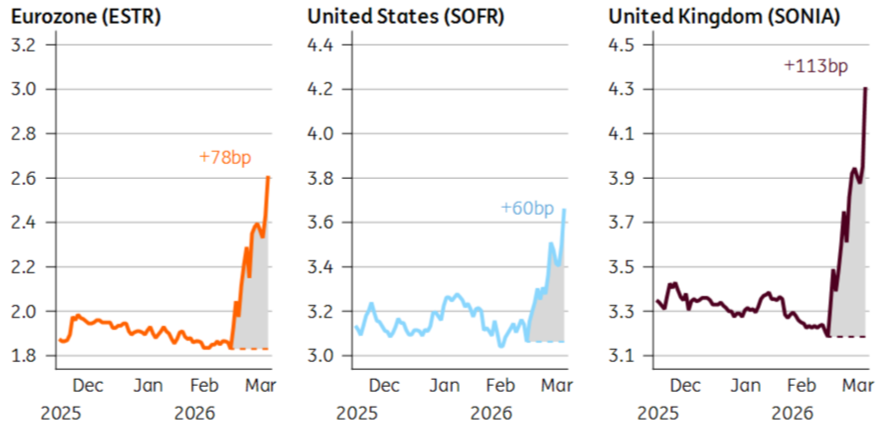
Why rate hikes risk winning the wrong inflation fight

This was the week when central banks gave an inch on rate hikes, and investors took a mile. As far as markets are concerned, Pandora's box is open; the path to higher interest rates is set.

Just look at the chart. One year from now, interest rates are priced more than half a percentage point higher than they were pre-war in both the US and the eurozone – and considerably higher still in the UK. Three hikes or more are priced from both the ECB and the Bank of England by year-end.

Markets have repriced interest rates dramatically higher

Implied one-month rate in one year (1Y1M swap)
Change since 27 February



Source: Macrobond, ING

Data as of 19 March close

That says as much about markets as it does central banks. Positioning and liquidity are exaggerating the story. My Rates Strategy colleagues caution against taking market pricing entirely at face value.

Yet central banks were unquestionably hawkish this week. And it didn't have to be that way.

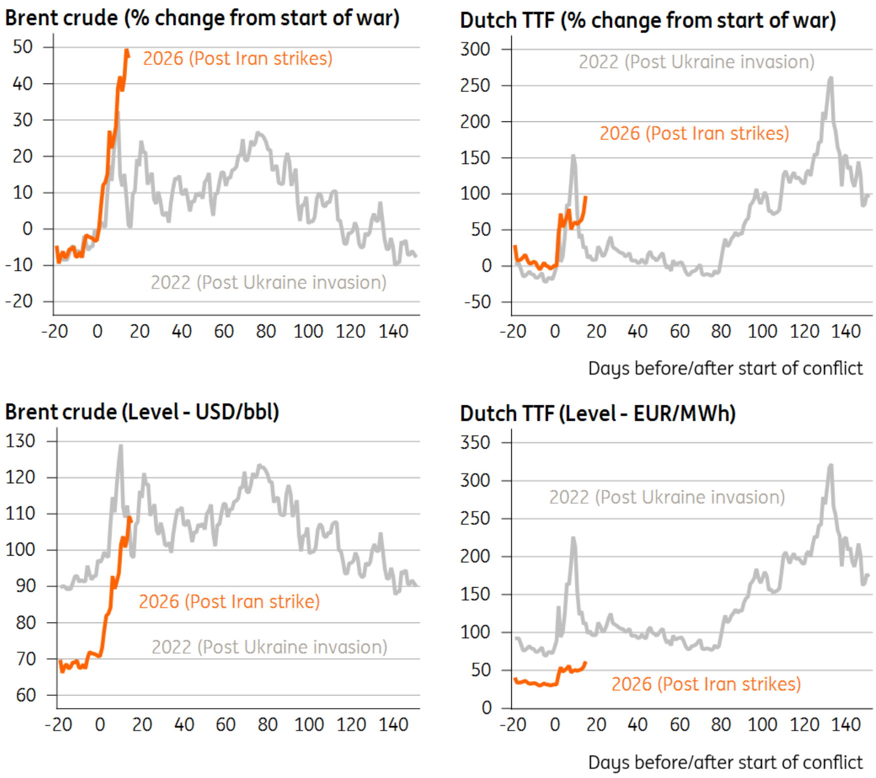
Officials could have simply said that the situation is uncertain, that nobody knows how long the disruption will last, that they're being vigilant and will see where the dust settles by the next meeting. And left it at that.

But policymakers went out of their way to burnish their rate-hiking credentials. Fed Chair Jerome Powell expressed his mounting frustration that inflation has been above target for five years now. In Britain, even the arch-doves, who until a month ago were ardent rate-cutters, were out floating the possibility of rate increases. And over in Frankfurt, the ECB hawks are on the airwaves actively talking up the chances of an April hike.

Above all, this tells us the painful memories of the last inflation spike are playing a key role in policymaking right now. Speak to many central bankers, and they will tell you both that they were too late to hike rates back then, and too quick to cut them when inflation started coming down.

That is not a mistake they want to make again. And to be fair to them, the percentage change in energy prices has now eclipsed the immediate fallout of the 2022 Ukraine invasion. That is what matters for inflation, even if, in level terms, things aren't looking quite so dramatic, at least on natural gas. The missile strike against Qatari LNG facilities – and the reportedly lengthy repair times involved – may well have influenced the tone of the ECB/BoE decisions.

2022 vs 2026: How the spike in energy prices compares



Source: Macrobond, ING

The central question that policymakers need to answer is this: *Will higher energy prices drive up core inflation?* That is what will determine whether they can "look through" the Middle East crisis or not.

Evidently, one big concern among officials is that the higher prices go, and the longer they stay high, the more likely it is that the inflationary impact snowballs. The relationship between energy prices and areas like food and services inflation is not necessarily linear.

Firms can weather short-term price fluctuations in their margins, but not forever. And the higher inflation goes through the second half of the year, the greater the impact on next year's annual price resets. Think of your internet or phone bills, for instance, that might explicitly be indexed to prior rates of inflation. Something similar is true of collective bargaining discussions in Europe.

That kind of thing risks making the whole inflationary episode longer-lasting – and if you listen to Bank of England Chief Economist Huw Pill, we're probably close to the inflection point. He's previously argued that when inflation hits 4%, it's statistically much more likely to stay high for a prolonged period. And by my reckoning, if energy prices are sustained at today's levels (a big if), it just about gets us there.

That's the fear, anyway. Personally, I just don't buy it.

Central banks are concerned about rising inflation expectations. And rise, they will. Ask any consumer where they think inflation is going, when they can see their petrol prices rising, and surprise! They will tell you it is going up.

But that makes no difference whatsoever if consumers (and businesses, for that matter) can't act on that information. In an environment where vacancy rates are dramatically lower than when the last energy shock hit, workers simply don't have the same power to seek faster pay rises. Nor, in many cases, do companies possess the power to materially pass on higher energy costs, either.

If that's the case, then those firms are much more likely to deal with higher costs by cutting staff. And remember, the whole "non-linear" argument applies just as much to the jobs market as it does inflation. Job losses, once they begin at scale, have a habit of spiralling as economic demand weakens across the economy.

Something similar is true in financial markets: the higher energy prices and interest rate expectations go, the greater the risk that something breaks. And when that happens, the ripple effects can often be felt far and wide.

Central banks are naturally well aware of this – and said as much this week. The BoE, for instance, said explicitly that higher energy prices could necessitate rate cuts if the fallout is primarily through the jobs market.

Still, that won't stop central banks talking up the possibility of rate hikes over the coming days, in the hope words can do some of the work for them. Some of this is directed squarely at governments. ECB President Lagarde explicitly warned them to keep energy support "*temporary, targeted and tailored*". The message is clear: government support, of the likes we saw in 2022, will build the case for rate hikes. Spain is the first out of the traps today with VAT cuts on electricity bills.

Regardless of that, there may come a point, if this crisis worsens, where central banks feel they have no choice but to preserve their inflation-fighting credentials and tighten policy.

In our team's more extreme energy price scenario, where oil averages 120 USD/bbl in Q2, we'd expect the ECB to hike twice this year. The same is probably true of the BoE. Markets are certainly right about one thing: if the central banks do raise rates, they are unlikely to do so only once.

But it's a risky strategy. Central banks rightly need to avoid a re-run of the previous inflation overshoot. But in doing so, they risk fighting the last battle.

It's one thing to talk about rate hikes, quite another to deliver them. They are not our base case for the Fed, ECB or BoE just yet.

Ultimately, this is not 2022. The economy looks markedly different. And rate hikes may only compound the economic challenges coming down the track.

James Smith

THINK Ahead in developed markets

United States (James Knightley)

- It is a very quiet week for scheduled data and events, but there will be plenty of Fed officials speaking to the media. Their "dot plot" continues to point to a 25bp rate cut this year and a further cut next year, but markets are increasingly concerned about stagflation risks from higher energy costs and ongoing supply distribution issues through the Straits of Hormuz.

We argue that the US is insulated from this to some extent relative to Asia and Europe, but is not immune to the issues. The longer the blockages last, the greater the concern about fertiliser, food and plastics prices.

- That said the Fed has a dual mandate - preserving price stability and maximising employment - and the second part is also facing greater challenges. If the jobs market was stalled when the economy was looking in decent shape before the Middle East conflict started, an overlay of heightened geopolitical, economic and market angst is not going to incentivise business to suddenly start hiring now. Hence why we still feel the Fed is more likely to cut than hike rates and that is the messaging we expect to hear from Fed officials.

United Kingdom (James Smith)

- **Inflation (Wed):** February's data, which is likely to show headline CPI unchanged and services inflation dipping back, is largely irrelevant at this point. Nor is the fact that inflation is set to dip close to 2% by April, given that it will take until the third quarter to see the full effect of higher natural gas prices on electricity bills. At this stage, we think the BoE is headed for a prolonged pause.

THINK Ahead in Central and Eastern Europe

Poland (Adam Antoniak)

- **Retail sales (Mon):** Retail sales remain the bright spot of Poland's economy in early 2026. While industrial production was stagnant and construction nose-dived amid harsh weather conditions in the first two months of this year, the cold winter even boosted garment sales in January. We believe that the positive trends in durable goods demand observed in recent quarters were sustained in February. March should indicate how resilient consumer demand is to uncertainty and higher gasoline prices triggered by the outbreak of military conflict in Iran. Deterioration in March consumer confidence indicators suggests that this new shock may pose some risks to consumption ahead.

Hungary (Peter Virovacz)

- **Rate setting meeting** (Tue): Governor Varga's words from the February NBH rate setting meeting, 'this is not a rate cut cycle', aged pretty well. Eventually, it has aged too well. On 24 March, we saw the NBH turn into a hawkish currency defender, eventually closing the door on easing for now, but citing maximum flexibility to keep inflation on track and calm investors. We now anticipate only one more rate cut in 2026 as our base case scenario. If our 'long war' alternative scenario materialises, we see no room for a rate cut this year.

Czech Republic (David Havrlant)

- **Confidence** (Tue): Czech consumer confidence likely weakened in March, continuing its descent from last year's November peak. That said, the index remained at a relatively optimistic level, supported by lower energy bills and solid economic growth. However, the rising unemployment rate and the onset of the Middle Eastern conflict weigh on consumer sentiment. The outlook of higher oil and natural gas prices, along with potential disruptions to supply chains, also cooled the business sentiment in March, despite the recent stabilisation in manufacturing.

Key events in developed markets next week

Country	Time	Data/event	ING	Prev.
		Monday 23 March		
Eurozone	1500	Mar Consumer Confidence Flash	-	-12.2
		Tuesday 24 March		
US	1215	ADP Employment Change Weekly (000s)	10	9
	1345	Mar S&P Global Manufacturing PMI Flash	-	51.6
	1345	Mar S&P Global Services PMI Flash	-	51.7
	1345	Mar S&P Global Composite PMI Flash	-	51.9
Eurozone	0900	Mar HCOB Manufacturing PMI Flash	-	50.8
	0900	Mar HCOB Services PMI Flash	-	51.9
	0900	Mar HCOB Composite PMI Flash	-	51.9
Germany	0830	Mar HCOB Manufacturing PMI Flash	49.5	50.9
	0830	Mar HCOB Service PMI Flash	52.5	53.5
	0830	Mar HCOB Composite PMI Flash	52	53.2
France	0815	Mar HCOB Composite PMI Flash	-	49.9
UK	0930	Mar S&P Global Composite PMI Flash	-	53.7
	0930	Mar S&P Global Manufacturing PMI Flash	-	51.7
	0930	Mar S&P Global Services PMI Flash	-	53.9
		Wednesday 25 March		
US	1230	Q4 Current Account (USD bn)	-310	-226.4
Germany	0900	Mar Ifo Business Climate	87.5	88.6
	0900	Mar Ifo Current Conditions	87	86.7
	0900	Mar Ifo Expectations	88.5	90.5
UK	0700	Feb Core CPI (YoY%)	3.2	3.1
	0700	Feb Services CPI (YoY%)	4.2	4.4
	0700	Feb CPI (MoM%/YoY%)	0.4/3.0	-0.5/3.0
		Thursday 26 March		
US	1230	W 1 Initial Jobless Claims (000s)	215	205
Eurozone	0900	Feb Money Supply (YoY%)	-	3.3
Germany	0700	Apr GfK Consumer Sentiment	-25.6	-24.7
Italy	0900	Mar Consumer Confidence	96.8	97.4
Spain	0800	Q4 GDP (QoQ%/YoY%)	-/-	0.8/2.6
Norway	0900	Key Policy Rate	-	4.00
		Friday 27 March		
US	1400	Mar Michigan Sentiment Final	53	55.5
UK	0700	Feb Retail Sales (MoM%/YoY%)	-/-	1.8/4.5
Spain	0800	Mar CPI Flash NSA (MoM%/YoY%)	-/-	0.4/2.3

Source: Refinitiv, ING

Key events in EMEA next week

Country	Time (BST)	Data/event	ING	Prev.
		Monday 23 March		
Poland	0900	Feb Retail Sales (YoY%)	7.8	4.4
	1300	Feb M3 Money Supply (YoY%)	9.6	10
		Tuesday 24 March		
Poland	0900	Feb Unemployment Rate	6.1	6
Czech Rep.	800	Mar Consumer Confidence	105.2	107.6
	800	Mar Business Confidence	98.2	99.8
	800	Mar Composite Confidence	99.4	101.1
Hungary	1300	Mar Hungary Base Rate	6.25	6.25
		Wednesday 25 March		
Russia	1200	Feb Industrial Output	2.8	-0.8
		Friday 27 March		
Hungary	0730	Feb Unemployment Rate	4.7	4.6
	0730	Q4 Current Account (EUR bn)	-0.14	0.9

Source: Refinitiv, ING

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Transformative or overhyped? The impact of weight-loss drugs on European food demand

The global market for GLP-1s will reach $100bn in 2027. Yet, the impact of these drugs on food demand is still limited in Europe. Around 2% of European adults are using GLP-1s, but that number will grow. Food makers face a gradual demand shift, giving them time to respond by changing products and marketing and investing in markets with less GLP-1 use

The move towards oral GLP‑1s could reshape demand in the years ahead

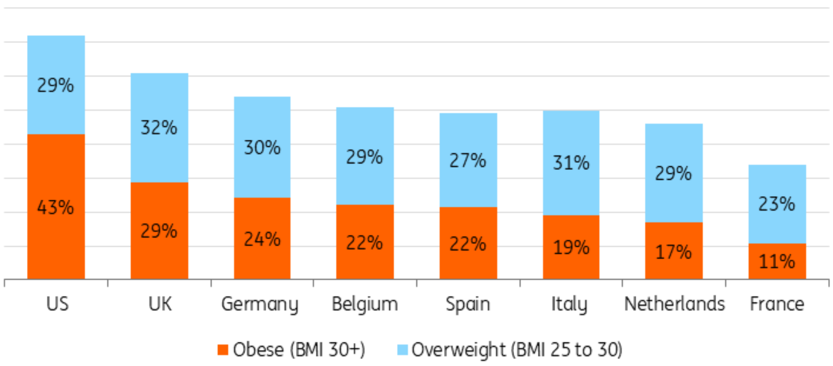
Obesity and Type 2 diabetes are significant societal and economic problems

Obesity rates have risen sharply over the past few decades: in 1975, a little over 10% of US adults were obese; in 2022, 43% of US adults were obese. In Europe, obesity rates are lower but still considerable. In fact, there are currently more people overweight than malnourished worldwide. Weight loss drugs (or GLP-1s) are very effective in both the treatment of diabetes and for weight reduction (users typically lose 12-22% of body weight). No wonder everyone is looking at these blockbuster drugs as a powerful tool to reverse rising obesity rates.

This is not just important because it helps people lead healthier lives, but obesity also carries high societal costs. It is linked to costly health issues such as cardiovascular diseases, joint problems and diabetes. Additionally, obesity reduces labour force participation and lowers productivity, as well as impacting physical and mental well-being.

Obesity rates are lower in Europe than in the US and vary widely within Europe

Share of adults, 2022

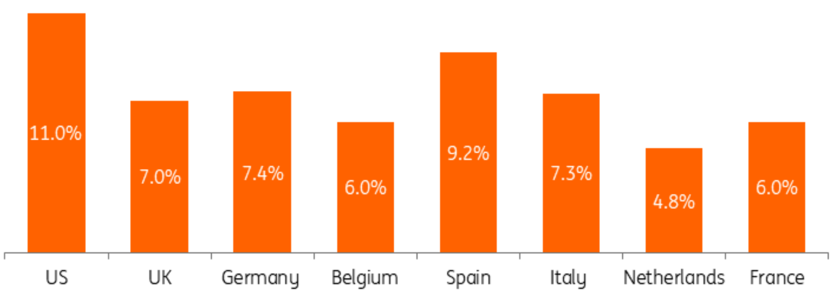


Source: World Health Organization, Global Health Observatory

Prolonged obesity can increase the risk of diabetes significantly. Roughly 90% of all diabetes patients have Type 2 diabetes, which means that they develop diabetes over their lifetime rather than being born with it. Genetics, lifestyle, and weight are key factors in developing Type 2 diabetes. This makes the case for GLP-1s even stronger, because diabetes takes a significant toll on patients and is a chronic disease with very costly complications (such as kidney dialysis or cardiovascular diseases).

Type 2 diabetes is a common chronic disease in Europe and the US

Share of adults with Type 2 diabetes, 2024



Source: International Diabetes Federation, ING Research

Use of weight-loss drugs is growing quickly

With that in mind, it's no surprise that GLP-1s have seen a very steep growth trajectory since they were introduced. In 2025, the global market size for GLP-1s was roughly $70bn, which includes treatment for diabetes and weight loss. Roughly one in five US adults has used a weight-loss drug, and current use is estimated to be around 12%.

In Europe and the UK, we estimate that at least 2% of adults are currently using them, which means about nine to 10 million users. Our estimate considers a variety of sources, including data from public health authorities, consumer surveys and revenue data from Novo Nordisk and Eli Lily. Still, it's likely to be conservative, as actual usage precedes hard data.

In Europe, weight-loss drugs have been adopted more quickly in wealthier countries like the UK and the Nordics. Central and Eastern European countries show slower uptake even though they're among the countries with the highest obesity rates. There is a great variety of users. You'll have people who pay out-of-pocket to use them for weight management and people who get them as a treatment for Type 2 diabetes. In terms of age and gender, older consumers and women are over-represented.

A lot of people don't stick to the drugs. Studies show that 50% discontinue GLP-1s within one year for various reasons, including costs, side effects or simply because they've reached a certain target weight. That is important for food and beverage companies that could see some of their products falling temporarily in and out of favour with consumers as a result.

12% vs 2%

Use of GLP-1s amongst adults

US vs EU+UK

Market size to reach $100bn in 2027

The trillion-dollar question is: how will demand evolve over the next few years? As mentioned, the market for GLP-1s was $70bn last year. Several factors will determine the future market size.

First, the GLP-1 market will become predominantly oral in the US in 2026 and in Europe in 2027. This means lower prices (up to 50%) and likely higher use. Second, more companies will launch their own weight loss drugs, which means more competition and price pressure. Third, governments will consider reimbursing costs for more patients if weight loss is sustainable and helps with prevention, which would further increase demand. Considering that revenues doubled for many weight-loss drugs in 2025, the size of the problem and the fact that only 40% of people living with diabetes receive treatment (Novo Nordisk), we estimate that the market will grow by 20% in the next two years. That means it would hit $100bn by the end of 2027.

Yet, this still leaves the biggest question unanswered: how sustainable is the weight loss achieved with GLP-1s? Current academic research indicates that patients gradually regain weight when they stop using GLP-1s. It is important to note that this often happens with people who achieve significant weight loss, as obesity is a complex disease. The final verdict has not been cast, especially as newer therapies are being developed and the use of GLP-1s for weight loss is still in its infancy in Europe.

Very limited impact on food and beverage demand in 2026

Many argue that GLP-1s can have a transformative impact on aspects of modern life, including our food consumption habits. Questions on the impact do pop up during earnings calls of major food makers, but the drugs are not viewed as an immediate threat. That makes sense; different studies indicate that GLP-1 users reduce their calorie intake by 15-20%. So, if 2% of all adults in the EU and UK use them, then the current impact on total calorie demand is about 0.25%. That's not a worrying number for food and beverage companies yet.

While GLP-1 use is growing fast, it's happening from a low base, so we don't anticipate a massive blow to demand in 2026 and 2027. Still, it shouldn't be neglected. There is already little volume growth in the food market in Europe. Studies in the US point out that expenditure and intake of categories like savoury snacks, candy, chocolate and alcoholic beverages are more exposed (see here and here).

So, if GLP-1 adoption goes up, it can become a drag on growth potential for companies operating in affected categories. In response to changes in demand, bakery company Grupo Bimbo has put more emphasis on reformulation, portion control and sugar reduction, while The Magnum Ice Cream Company has detected a shift to fewer but more premium treats. Eventually, changes in consumption also trickle down to ingredient suppliers. As an example, some sugar traders have reduced their sugar consumption figures for North America and Europe as a result.

Households cut spending on most food and beverage items when a member uses GLP-1s

Changes in grocery spending of US households on a selection of categories, 6 months post GLP-1 adoption

Change in grocery spending	Category
>10% reduction	Chips & savoury snacks
5-10% reduction	Sweet bakery, cheese, meat
2.5-5% reduction	Fresh vegetables, soft drinks, milk, coffee/tea/energy drinks, candy & chocolate
0-2.5% reduction	Alcohol
0-2.5% increase	Fresh fruit

Source: Hristakeva, Sylvia and Liaukonyte, Jura and Feler, Leo, The No-Hunger Games: How GLP-1 Medication Adoption is Changing Consumer Food Demand.

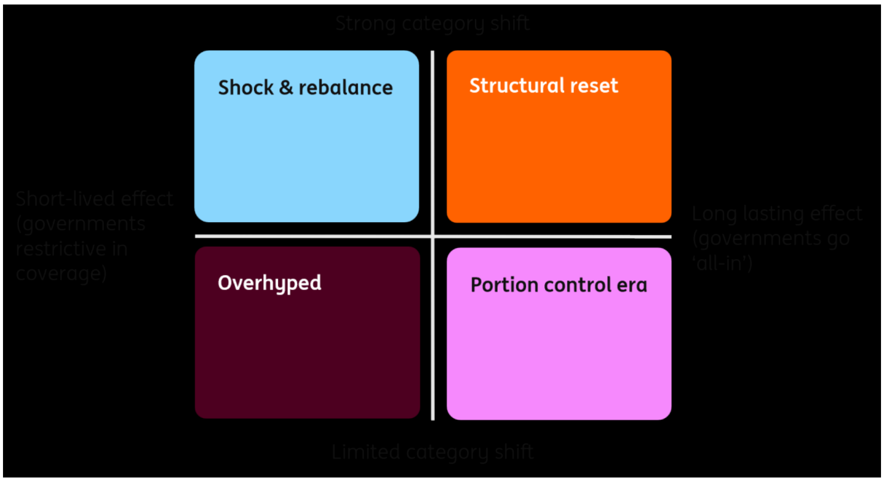
Transformative or overhyped? The potential impact by 2030

Despite increasing use and their possible economic benefits, much is still uncertain about the longer-term impact of weight-loss drugs on the food and beverage market. To provide senior executives with a framework, we propose four scenarios in which we differentiate between our two biggest questions. The first is the duration of the effect on weight, calorie intake and body weight.

If there is a sustained long-term effect, we expect that governments will be more inclined to rapidly expand coverage. If not, then governments remain restrictive.

The second dimension is the strength of any shifts between categories. On one side of the spectrum, we'll see that GLP-1 users consume less of everything; on the other side, the reductions are very concentrated in only a few categories or even restricted to certain products in these categories.

Our four scenarios for the GLP-1 impact on the food & beverage industry



Source: ING Research

Key take-aways from our scenarios

Scenario	Narrative	2030 penetration (% of adult population)	Volume impact (% change in total calorie intake)	Category impact	Strategic takeaway for food & beverage companies
Shock & rebalance	Demand rises sharply in 2026–27, then stabilises as growth normalises.	8–12%	-1.25 to -1.75%	Concentrated in a few categories	Disproportionate volume pressure in most exposed categories; opportunities in nutrient-rich alternatives
Structural reset	GLP-1s become a mainstream health and prevention tool, supported by increased public spending on compensation.	15-20%	-2.5 to -3.5%	Widespread across categories	Companies must adapt to structurally lower demand; strong case for reformulation and nutrient density.
Overhyped	Initial excitement fades as long-term effects and willingness to stay on treatment disappoint.	3–5%	-0.25% to -0.5%	Concentrated in a few categories	Too small to materially reshape the sector. Limited need for product innovation and dedicated products.
Portion control era	Effects prove durable enough to matter, but not transformative enough to drive a prevention-led healthcare shift.	8-12%	-1.25 to -1.75%	Widespread across categories	Portion downsizing and reformulation become a central theme for a major part of the industry.

Source: ING Research

Calorie intake could drop by 2.5-3.5% in 2030 in a transformative scenario

In our most transformative scenario, increased use of GLP-1s has the potential to shave off 2.5-3.5% of total calorie intake in 2030 and higher percentages for the most affected categories. To make it more tangible, many GLP-1 users reduce their daily intake by 700 calories or more. That is equivalent to two cheeseburgers at a fast food restaurant or three-quarters of a 200-gram bag of crisps. If the industry fails to adapt, that brings a risk of overcapacity in production.

In our overhyped scenario, the impact on demand remains very limited because usage of the drugs remains restricted to the most pressing cases and occasional use for weight loss. In this scenario, we'll see a reality check once initial success stories fade and scepticism rises.

The most likely outcome is that we see a combination of the scenarios across European countries. That's because countries: 1. have different public health systems, 2. face different levels of constraint on public health budgets, and 3. cope with varying levels of Type 2 diabetes and obesity prevalence. It's clear that things are changing within national health authorities; the NHS in England approved the use of Mounjaro® for certain patients in 2025, the HAS in France now allows general practitioners to prescribe the drugs, and Dutch authorities are considering adding more weight-loss drugs to the basic health package. But authorities are proceeding with caution because costs precede benefits. For European food and beverage makers, keeping track of such decisions will help to understand where the market is headed.

Food and beverage companies will adapt to offset any impact

Unlike many recent supply chain shocks, companies can prepare for the increasing adoption of GLP-1s. We recommend considering the following:

- **Product.** Product innovation, including reformulation, smaller portions and premiumisation, appears as a key response. Corporates including PepsiCo, Mondelez, Hershey and Danone have all highlighted in recent earnings calls that innovation is an important part of their adaptation strategy. They also like to frame it as an opportunity to offer new products, including high-protein and fibre-enriched versions. In the mid to long-term, increasing a company's presence in less affected products and categories (such as healthy snacking) through organic growth or M&A also fits this strategy.
- **Marketing.** When more people use GLP-1s, it makes sense for impacted companies to redesign marketing strategies. You would expect branded food and beverage makers to focus more on consumers who are not using GLP-1s, which is most of the population in any scenario. This includes consumers who could benefit from medication but don't have access to it. Then there is also a drive to re-engage with consumers as soon as they come off GLP1s. After all, the temptations are still there, making it hard for consumers to sustain any changes in lifestyle and consumption habits. From a public health perspective, that is a bit of an inconvenient truth, but that's how markets work.
- **Geographic shift in sales markets.** By increasing focus on regions and countries with lower usage of GLP1s. For European food and beverage makers that pursue volume growth, it will further increase the need to turn to growth markets in Asia, Africa and Latin America. However, GLP-1s will start making inroads in some of these markets too. Novo Nordisk's semaglutide patent is set to expire in several countries, including China, India and Brazil, and pharmaceutical companies are lining up to produce more affordable generic variations. So, for any company considering a shift in focus, it's a sensible thing to gain a better understanding of the dynamics around weight-loss drugs in these markets.

GLP-1s provide a solution at individual level, but they won't fix the underlying problem in the food system

GLP-1s are a technological fix for problems related to obesity and Type 2 diabetes, but these diseases are symptoms of a systemic problem. The underlying issue of a food system geared towards convenience and overconsumption remains. Rapid growth in GLP-1 usage shows they're an attractive solution at the individual level. However, it will require a lot more effort to get from a short-term solution at the individual level to a lasting solution at the public health level.

The impact on food and beverage demand would be more profound if policymakers grasped the GLP-1 opportunity by taking additional steps to steer the food system in a healthier direction by using a combination of prevention measures, increased taxation of unhealthy foods, subsidies for healthier alternatives and marketing restrictions. Thus far, we don't see European policymakers seizing that opportunity, also because such measures are not very popular and often viewed as overprotective.

Meanwhile, for food and beverage makers, a gradual adaptation strategy will likely be sufficient for 2026 and 2027. Nevertheless, it's wise to consider more extreme scenarios to avoid surprises a few years from now.

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FX Daily: Hawkish hangover

Powell's mildly hawkish hints on Wednesday were dwarfed yesterday by reports the ECB is considering a hike in April and the most dovish BoE member openly discussing monetary tightening. The dollar is suffering on the back of surprising overseas hawkishness, but also from some tentative optimism in energy markets

Bank of England Governor, Andrew Bailey. The Bank surprised on the hawkish side yesterday with a 9-0 vote in favour of a hold

USD: Fed sounds dovish relative to ECB and BoE

Oil had another wild ride yesterday, with Brent touching 119$/b before easing back closer to 107$/b. The correction was likely a function of Israel claiming that it is helping the US re-open the Straight of Hormuz, with Prime Minister Benjamin Netanyahu saying he sees "the war ending a lot faster than people think". That follows US discontent over Israeli strikes on Iranian gas fields earlier this week.

While oil prices remain elevated, the size of the dollar decline in the past 24 hours seems to embed some optimism about the war. What also contributed to USD losses were the hawkish surprises by the European Central Bank and Bank of England (discussed in the sections below), which seemed to dwarf the hawkish vibes from US Federal Reserve Chair Jerome Powell's press conference on Wednesday.

That said, our takeaway from this week of central bank decisions hasn't changed: not enough guidance has been offered to dent oil's role as a major market driver. Rate expectations should remain fluid and commodity price dependent, and continue to play a secondary role for FX.

The next few days will tell us whether this wave of cautious optimism has legs. The dollar can fall more on military de-escalation news, but clarity on the Straight of Hormuz reopening is necessary to prevent USD rebounds at a second stage.

Francesco Pesole

EUR: April on the table

The ECB opted for a cautious tone in light of energy price volatility yesterday, but President Christine Lagarde's press conference had a hawkish undertone as she conveyed a sense of heightened concern for upside risks to inflation. But even more importantly, Bloomberg later reported that ECB officials are already considering a rate hike in April should inflation rise too far above target.

That can be a game-changer. Despite the recent hawkish repricing, markets were pricing in a hike only from June before yesterday. The reasoning was that the ECB would have needed a couple of months of data to assess second round effects. Putting April on the table (now 15bp priced in) means that the ECB may be ready to act aggressively and pre-emptively, intuitively raising the chances of back-to-back increases.

Our economists aren't ready to pencil in a rate hike yet as a positive turn in the war and energy prices can still discourage the hawks. But the chances of a hike have undoubtedly increased, which raises the upside potential for the euro beyond the near-term impact of energy prices.

Speaking of which, gas prices have eased back after yesterday's spike, but Iran's attacks on Qatar's LNG facilities are now estimated to cause a loss of 12.8m tons per year (roughly 3% of global production) for three to five years. Further EUR gains from here depend on no additional major shocks to gas supply.

We think caution is very much warranted in EUR/USD at this moment, as the pair seems to be trading a bit too strong considering where oil and gas prices are. But should we see some de-escalation over the weekend, EUR/USD could be eyeing 1.170 soon, backed by a hawkish ECB message.

Francesco Pesole

GBP: Hawkish BoE surprise

The Bank of England surprised on the hawkish side with a 9-0 vote in favour of a hold, as consensus was that two members would still vote for a cut. Incidentally, the MPC's most dovish member, Swati Dhingra, openly discussed a rate hike to "stabilise rate setting dynamics" in case of a sharp rise in inflation.

The key takeaway from the statement was the BoE is "ready to act" against an inflation spike, and markets added a mammoth 50bp of hikes by year-end, for a total of 70bp. While it's often the case that hikes come in couples, we still believe market pricing is too aggressive, considering much less favourable conditions for second round effects relative to 2022.

The repricing of both ECB and BoE rate expectations has kept EUR/GBP relatively stable, with all the action happening in GBP/USD. Oil prices are now back in the drivers' seat, but the hawkish BoE tilt is offering some extra support.

Francesco Pesole

CZK: CNB not in rush despite risks

The Czech National Bank, as expected, left rates unchanged at 3.50%. However, forward guidance was where attention was focused yesterday. We heard a fairly balanced message from the governor, and the central bank sees itself in a comfortable situation compared to 2022 due to below-target inflation and real rates remaining positive. However, if core inflation were to increase due to rising energy prices, the central bank would increase rates, according to the governor. At the same time, if the negative impact on the economy were to grow, it would lead to rate cuts.

Of course, at this moment we cannot expect a strong view on further developments. For us, though, there was an interesting comment regarding this year's inflation, which should be below 2% according to the updated CNB forecast. While this cannot be ruled out, we see significant upside risk here and the CNB may be surprised quickly.

The market may have seen some calming in rate hike bets after the CNB meeting. However, the reaction may have been lost in global volatility, and we do not take much away from yesterday's meeting. EUR/CZK has been the least volatile pair in the region for now and that is likely to continue. FX should limit any reaction from the CNB, and we believe that the central bank will potentially be the last to hike rates in the CEE region.

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Rates Spark: Pushed higher by hawkish shifts

Front-end rates continue to push higher after a surprisingly hawkish BoE and reports that some at the ECB are already prepared to act in April. In EUR, longer rates are starting to struggle to keep up with front-end rates. But it does not look like a tipping point yet

The European Central Bank and Bank of England were seen as hawkish with yesterday's rate decisions, while acknowledging higher inflation expectations

Aggressive front-end pricing might overstate how quickly the European Central Bank can act

The impression that yesterday's European Central Bank meeting left – given the magnitude of events – was a relatively balanced one amid this environment of high uncertainty. Unsurprisingly, there was an acknowledgement of the increased upside inflation risks, but also of the downside risks to growth. Importantly, without really passing judgement as to what will hold more weight at this juncture.

There is a clear readiness to act, however, which itself is a hawkish shift in the ECB's stance. And President Christine Lagarde highlighted that the ECB will monitor everything from supply bottlenecks and firms' price-setting intentions to wage developments to determine the pass-through of the shock. Lagarde stood by a data-dependent and meeting-by-meeting approach to assess the ECB's appropriate policy stance.

While the ECB did not seem to signal imminent action at its meeting, subsequent press reports suggested that some officials were ready to raise rates as soon as April. The market is now fully discounting two hikes and the possibility of more this year. The upcoming April meeting has an implied probability of a 25bp hike of over 60%. But at the same time, markets do not set prices in a vacuum, and we saw, for instance, that just prior to the ECB meeting, markets digested a surprisingly hawkish Bank of England that also dragged up EUR rates – highlighting again that there is some element of positioning in play as well, and that market pricing should not be taken entirely at face value.

For now, oil prices and geopolitical developments will remain in the driving seat. That is until we reach a tipping point where adverse risk sentiment takes over. What we did observe was that the long end has had a greater struggle to keep pace with the rise in short-end rates. While the 2y swap rate has settled 15bp higher on the day, the 10y swap rate only got past the 3% mark briefly. 30y rates are outright lower on rising front-end rates for the past two sessions, accelerating the common flattening dynamic seen when front-end rates rise.

The Bank of England sees the strongest hawkish shift

Markets see the Bank of England as the hawkish one, adding a full 25bp hike this year in reaction to Thursday's meeting. That brings the total pricing for hikes to more than 60bp for 2026. We also saw strong moves in the 1y rate in 2y forwards, reflecting the idea that the BoE may keep policy tight for a more prolonged period. Even a dovish interpretation of the ECB meeting immediately thereafter did relatively little to bring the 2y GBP swap rate down again.

The interesting takeaway from this is that markets are happy to price in different reaction functions per central bank. One could say that central banks face a similar supply shock while the demand backdrop is relatively weak, which in a simplified world would argue for similar monetary policy actions.

But if we look at the UK in particular, the inflation backdrop does add complexity. The ECB has already demonstrated the willingness and ability to bring back inflation to a 2% target while the BoE still faces CPI figures closer to 3%. The risk of seeing unanchored inflation expectations could therefore be higher in the UK. We don't see this so far, however. The long-term GBP inflation swaps remain very well-behaved and are less than 10bp above February's numbers.

Friday's events and market view

We have a relatively light calendar in terms of data for Friday, which means we could have more focus on comments from central bankers after this week's meetings. Isabel Schnabel, for one, is scheduled to speak in the evening about the German economy, but we can expect more ad-hoc comments from others.

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Asia week ahead: Japanese inflation and South Korean sentiment data

Japan releases February inflation data, while South Korea reports on March sentiment. Other key releases include Taiwan's industrial production data

Asia Research highlights of the week

Asia FX Talking: Relatively broad-based pressure
China beats dreary expectations, but more work needed to support growth
Bank of Japan holds rates and offers no signal about its next move

Japan: CPI expected to cool further

Consumer price inflation is expected to slow further in February, primarily due to the government's energy subsidy. It's likely to show that the deceleration in inflation we saw before the Middle East crisis persisted. Yet core-core inflation is expected to stay well above 2.0%. Higher energy prices are likely to push up inflation again in the coming months, prompting the Bank of Japan to keep the door open to rate hikes. At the same time, the flash purchasing managers' index is expected to drop sharply, with manufacturing falling more than services amid concerns about events in the Middle East. Overall, the timing of the next hike remains uncertain. In our view, an April move is unlikely, as the BoJ needs more time to assess the effects of rising energy prices on the economy.

South Korea: Sentiment data becoming less optimistic

Survey results on South Korean sentiment are likely to be less optimistic, as heightened geopolitical tensions reduce both confidence and economic activity. Higher gasoline prices and financial market instability are expected to significantly dent consumer sentiment. This will likely be reflected in the business survey as well.

China: A calm week ahead

It's a quiet week is ahead for China, with most of the key economic data already out for the month. We'll get industrial profits data for the first two months of the year on Friday. Markets will watch for any improvement from the sluggish 0.6% year-on-year growth rate in 2025.

Taiwan: Moderating industrial production following Lunar New Year

Taiwan releases its February industrial production data on Tuesday. We expect activity to moderate toward 13.3% YoY thanks to the Lunar New Year effect. But production should be quite robust to start the year amid strong external demand.

Key events in Asia next week

Country	Time (GMT+8)	Data/event	ING	Prev.
		Monday 23 March		
Japan	0730	Feb CPI (MoM%/YoY%)	-0.4/1.5	-0.1/1.5
	0730	Feb Core CPI (MoM%/YoY%)	-0.4/1.7	-0.2/2.0
	0830	S&P Global Manufacturing PMI Flash	49	53
	0830	S&P Global Services PMI Flash	52	53,8
	0830	S&P Global Composite PMI Flash	51	53,9
Singapore	1300	Feb Core CPI (YoY%)	-	1
	1300	Feb CPI (MoM%/YoY%)	-	-0.5/1.4
		Tuesday 24 March		
South Korea	0500	Feb PPI (MoM%/YoY%)	-	0.6/1.9
Australia	0600	Mar S&P Manufacturing PMI Flash	-	51
	0600	Mar S&P Services PMI Flash	-	52,8
	0600	Mar S&P Composite PMI Flash	-	52,4
Taiwan	1600	Feb Industrial Output (YoY%)	13,3	28,5
		Wednesday 25 March		
Japan	1300	Jan Leading Indicator Revised	-	2,1
South Korea	0500	Mar Consumer Sentiment Index	105	112,1
Australia	0830	Feb Inflation (MoM%/YoY%)	-/-	0.4/3.8
		Thursday 26 March		
Singapore	1300	Feb Manufacturing Output (MoM%/YoY%)	-/-	5.3/16.6
Taiwan	1600	Feb Unemployment Rate	-	3,4
		Friday 27 March		
Indonesia	1200	Feb M2 Money Supply (YoY%)	-	10
South Korea	0500	Mar Composite Business Survey Manufacturing	94	97,1
	0500	Mar Composite Business Survey Non-Manufacturing	89	92,2
Philippines	0900	Feb Imports (YoY%)	-	-3,1
	0900	Feb Exports (YoY%)	-	7,9
	0900	Feb Trade Balance (USD bn)	-	-4
Taiwan	-	Mar Consumer Confidence	-	66,6

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The Commodities Feed: LNG supply disruptions now a long-term problem as Iran hits Qatari facilities

The global LNG market is set to be even tighter than expected through at least 2027, following Iranian attacks on Qatari facilities

Energy – Qatari LNG set for multiyear disruption

European natural gas prices surged to an intraday high of EUR74/MWh yesterday. Though the front-month contract settled at EUR61.85/MWh, prices were still up 13% on the day. This comes as the market digests the impact from Iranian attacks on Qatari LNG infrastructure.

The global LNG market is set to be tighter than originally expected for the foreseeable future following Iranian attacks on Ras Laffen in Qatar. According to Qatar Energy, the attacks have impacted 17% of the site's LNG export capacity. This is equivalent to just over 17 bcm, around 3% of global LNG trade. It could take 3-5 years to bring this capacity back online. When you factor in this disruption, along with delays in the start-up of new export capacity from Qatar, it's no surprise that the TTF forward curve has moved quite a bit higher through 2027 just over the last day. The market is starting to price in a longer supply disruption. Also, the large global LNG surplus that the market had expected through 2027 is looking much more elusive.

ICE Brent eased from its intraday high of $119/bbl yesterday, with it trading back below $108/bbl. This comes after Israel said it would no longer target Iranian energy infrastructure. There are also suggestions that President Trump may ease some sanctions on Iranian oil to help limit price gains. Additional downward pressure reflects the US administration ruling out export restrictions on oil. There were suggestions of such a move. This added to the growing disconnect between Brent and WTI, with the Brent-WTI spread trading to $13/bbl, the widest level since 2014.

Iranian attacks on neighbouring energy infrastructure are also affecting the refined products market, which continues to strengthen. The prompt ICE gasoil timespread surged to a backwardation of $160/t, while the May gasoil crack surged above $45/bbl. The jet fuel market is also tightening, as reflected in the continued strength in the jet regrade. According to Insights Global, gasoil inventories in the ARA region fell by 105kt week-on-week to 2.09mt, while jet inventory fell by 42kt WoW to 737kt.

Metals - Metals sell off amid broader market rout

Metals were under heavy pressure on Thursday amid a broader selloff across global markets, as energy prices surged following strikes on energy infrastructure amid the escalation between Iran and Israel. Growing concerns over the global economic fallout from the conflict are weighing on risk appetite. The spike in oil prices has added to inflation concerns, reducing the likelihood of a near-term US rate cut and creating headwinds for both industrial and precious metals.

Aluminium was among the worst performers. It slumped more than 8% at one point – the sharpest intraday drop since 2018 – as fears of slowing global growth outweighed supply disruption risks. Copper fell more than 5%, dropping below $12,000/t at one stage, wiping out its gains for the year. Gold slid over 5% and silver plunged as much as 10% at one stage.

In China, stockpiles of both aluminium and copper have continued to build, reinforcing concerns about near term demand. Primary aluminium inventories on the SHFE surged to their highest level since 2020, adding to the bearish tone across the complex.

Gold has been under pressure from persistent ETF outflows in recent weeks. Investor demand has weakened after the latest Federal Reserve meeting dashed hopes of an imminent rate cut. Historically, ETF holdings tend to rise alongside prices and are closely linked to expectations for US monetary policy. While gold remains up around 6% year to date, upward momentum has faded. Some investors are selling gold to raise cash or rebalance portfolios.

Agriculture– Looser soybean and tighter corn market in 2026/27

In its initial 2026/27 projections, the International Grains Council expects global soybean production to rise from 426mt to 442mt, with consumption increasing to 442mt from 430mt. Ending stocks are expected to inch up to 79mt from 78mt. In contrast, global corn output is forecast to fall to 1,303mt (1.3% year-on-year), while consumption is expected to grow to 1,315mt (+0.9% YoY). This reduced ending stocks to 294mt from 306mt. Wheat production is projected to decline from 845mt to 822mt, while consumption rises to 829mt from 825mt, pushing ending stocks down to 276mt from 283mt.

Lower wheat and corn production outlooks are mainly driven by surging fertiliser prices. It's linked to supply disruptions in the Strait of Hormuz and shutdowns of regional production facilities. Although most Northern Hemisphere producers remain adequately supplied ahead of spring planting, a prolonged crisis could influence planting decisions later in the year. Fertiliser-dependent regions in Asia and Africa are particularly vulnerable. Prolonged supply issues may reduce fertiliser use—potentially pressuring yields and crop quality.

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ECB makes a hawkish pivot at its March meeting

Rates have been kept on hold, but the European Central Bank has clearly opted for a hawkish pivot at today's meeting

ECB President Christine Lagarde at today's press conference in Frankfurt

Comments by European Central Bank President Christine Lagarde at the press conference gave today's meeting a more hawkish tilt. Even if a rate hike is not imminent, the change in tone and language acknowledges more uncertainty and is meant to demonstrate the ECB's willingness to act, if need be.

The fact that the well-known 'monitor closely' or 'closely monitoring' is back is a clear signal that the ECB has shifted to higher alert. In the past, the term 'monitor closely' had always been a sign of high alertness; the time it was used was during the short-lived banking tensions in March 2023 and before in 2022. In the distant past, 'monitor closely' was followed by 'vigilance' in the run-up to rate hikes. Following the logic of institutional ECB language, today's 'closely monitoring' combined with Lagarde's statement that risks to the inflation outlook were tilted to the upside are both clear signals of increased alertness.

New round of forecasts

Just for the record – though clearly not really relevant for future policy decisions despite a later cut-off date than usual (March 11) – the latest round of ECB staff projections has GDP growth coming in at 0.9% in 2026, 1.3% in 2027 and 1.4% in 2028. Inflation is expected to come in at 2.6%, 2.0% and 2.1% over the next few years.

In this base case scenario, ECB staff treats the current oil price shock mainly as a one-off – and one that would hardly necessitate a monetary policy reaction. However, in two alternative scenarios ECB staff prepared, the monetary policy reaction could be different as both scenarios worsen the stagflationary impact.

In the adverse scenario, the impact on the economy would be temporary and part of the inflationary impacts unwind. Growth would be some 0.3ppt lower than the baseline in 2026 and 0.1ppt lower in 2027. Inflation would be 0.9ppt higher in 2026 but would come down quickly, pushing headline inflation 0.5ppt lower in 2028.

In the severe scenario, energy prices would have a stronger and longer-lasting effect, reducing GDP growth by 0.5ppt and 0.4ppt in 2026 and 2027 respectively. The eurozone economy would be in a technical recession in the summer of 2026. Inflation would also be much higher, e.g., 3ppt higher than the baseline in 2027.

Decoding the ECB's reaction function

We reflected on the ECB’s reaction function to oil price shocks and how it has changed over time here. During the press conference, Lagarde kept it vague, but reading between the lines suggested that the ECB is once again making a distinction between a supply-side and a demand-side shock. Stressing that the central bank would focus on commodity markets, potential supply bottlenecks, selling price expectations of firms, demand indicators and wage trackers, Lagarde gave the impression that the ECB would only start to react with rate hikes if and when higher energy prices and headline inflation would start to be passed through.

This is probably the most important lesson from the 2022 period: not that the ECB was too late to react to an energy price shock, but that it was too late to identify and to react to supply-driven inflation broadening to a fully-fledged and broad inflation problem.

On hold for now but rate hikes cannot be excluded

Looking ahead, an inflation wave is clearly in the making, from the direct impact of higher gas and oil prices on gasoline prices or retail energy prices next winter, to knock-on effects on transportation services, fertilisers and food prices. However, even a somewhat longer-lasting inflation overshoot does not have to be a concern. And, in fact, the only hurdle to calling this a typical supply side shock is the ECB’s own institutional memory, its credibility as an inflation fighter and the fact that ‘team transitory’ turned out to be very wrong in 2022. It’s obvious that ‘transitory’ has become a forbidden word at the ECB.

In a scenario in which the war in the Middle East and soaring energy prices remain limited in time, the ECB will talk like a hawk but not walk like a hawk – or, rather, fly like one. However, if energy prices stay high or higher for longer and find their way into other parts of the eurozone economy, the central bank apparently wouldn’t shy away from rate hikes. Even though the bar for a rate hike seems to be higher than markets have currently priced in, given that the starting position for the ECB is currently very different from in 2022 (i.e., interest rates are higher), inflation and inflation expectations are much lower than in 2022.

All in all, a rate hike is not yet on the table, but today's meeting clearly marks a hawkish pivot.

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ING's updated forecasts amid energy disruption

Our economists and strategists have updated their forecasts to reflect the rapidly changing events in the Middle East and our new base case scenario

Oil tankers and cargo ships are being forced to reroute as the Strait of Hormuz is effectively blocked

Interim forecast update

The war in the Middle East continues, and the Strait of Hormuz is practically blocked. Our initial base case assumptions from our March Monthly regarding the war have become outdated. We are normally not in the business of permanently changing our forecasts, but given that our next Monthly Update is only due in April, and we think that these times currently require some guidance, we have updated our full set of forecasts.

Three new scenarios

We have developed three new scenarios to guide our thinking and macro and market forecasts. In summary:

- **Scenario 1** (our new base case): Intensive combat will end within the next two weeks, but will be followed by lower intensity strikes that linger for several months. It's a scenario in which the opening of the Strait of Hormuz will take much longer than initially anticipated.

The 'end game' in this scenario could be a temporary ceasefire or tacit stand down with maritime escorts and exploratory talks on sanctions. Regime change over time is still possible, driven from the inside, as in East Germany in the late 1980s.

- **Scenario 2** (optimistic alternative): War ends surprisingly earlier than in our new base case scenario and the Strait of Hormuz opens up within a short period.
- **Scenario 3** (longer war): Longer combat action followed by protracted, lower-grade, impulsive confrontation. That would imply a much more uncertain situation in the Strait. In this scenario, the 'end game' could be a hard won ceasefire with third party guarantees (likely involving Russia or China), the possibility of sequenced sanctions relief, and a maritime security regime for Hormuz. However, even then, a durable peace could remain elusive, and the region might settle into a fragile, flare up prone equilibrium.

In short, our base case scenario does not foresee structural damage to an oil or gas production facility. We are facing a delay in energy and other shipments, but not (yet) a structural reduction. Needless to say, if a delay lasts long enough, it can still lead to these structural reductions. Generally speaking and for the time being, this is still mainly a supply-side shock, pushing up inflation and posing new challenges for central banks. As a result, the upward revision to our inflation forecasts is clearly larger than the downward revisions of growth, and they also differ across regions. What these new scenarios mean for our market forecasts can also be found here.

Contact us directly to discuss more details of the underlying scenarios and their implications.

ING Global Forecasts

	1Q26F	2Q26F	2026 3Q26F	4Q26F	2026F	1Q27F	2Q27F	2027 3Q27F	4Q27F	2027F
United States										
GDP (% QoQ, ann)	3.1	2.4	1.8	1.9	**2.5**	1.9	2.0	2.0	2.0	**2.0**
CPI headline (% YoY, avg)	2.5	3.8	3.5	3.2	**3.3**	2.8	2.0	1.9	2.1	**2.2**
Federal funds (%, eop)	3.75	3.75	3.50	3.25	**3.25**	3.25	3.25	3.25	3.25	**3.25**
3-month SOFR rate (%, eop)	3.70	3.70	3.50	3.30	**3.30**	3.30	3.30	3.30	3.30	**3.30**
10-year interest rate (%, eop)	4.20	4.30	4.20	4.15	**4.15**	4.15	4.25	4.25	4.30	**4.30**
Fiscal balance (% of GDP)					**-6**					**-6.1**
Gross public debt / GDP					**102.6**					**104.5**
Eurozone										
GDP (% QoQ, ann)	0.7	0.6	1.0	1.5	**0.9**	1.6	1.6	1.5	1.4	**1.4**
CPI headline (% YoY, avg)	2.0	3.2	3.1	2.7	**2.8**	2.6	1.9	1.8	2.0	**2.1**
ECB Deposit Rate (%, eop)	2.00	2.00	2.00	2.00	**2.00**	2.00	2.00	2.00	2.00	**2.00**
3-month interest rate (%, eop)	2.00	2.00	2.10	2.10	**2.10**	2.10	2.10	2.10	2.10	**2.10**
10-year interest rate (%, eop)	3.00	3.00	2.95	2.85	**2.85**	2.90	3.00	3.00	3.10	**3.10**
Fiscal balance (% of GDP)					**-3.6**					**-3.5**
Gross public debt/GDP					**93**					**93.9**
Japan										
GDP (% QoQ, ann)	1.2	0.8	1.2	1.6	**0.8**	0.4	1.2	0.8	0.8	**1.0**
CPI headline (% YoY, avg)	1.7	2.4	2.4	2.3	**2.2**	2.6	1.8	1.8	1.6	**2.0**
Target rate (%, eop)	0.75	1.00	1.00	1.00	**1.00**	1.25	1.25	1.25	1.50	**1.50**
3-month interest rate (%, eop)	1.20	1.20	1.20	1.35	**1.35**	1.50	1.50	1.70	1.75	**1.75**
10-year interest rate (%, eop)	2.25	2.50	2.50	2.60	**2.60**	2.75	2.75	2.85	2.85	**2.85**
Fiscal balance (% of GDP)					**-4.6**					**-5.0**
Gross public debt/GDP					**230**					**228**
China										
GDP (% YoY)	4.6	4.6	4.7	4.5	**4.6**	4.2	4.9	4.4	4.4	**4.5**
CPI headline (% YoY, avg)	0.9	1.1	1.4	1.3	**1.2**	1.6	1.1	1.1	1.1	**1.3**
7-day Reverse Repo Rate (% eop)	1.40	1.40	1.30	1.30	**1.30**	1.30	1.20	1.20	1.20	**1.20**
3M SHIBOR (% eop)	1.55	1.50	1.45	1.45	**1.45**	1.45	1.45	1.45	1.45	**1.45**
10-year T-bond yield (%, eop)	1.85	1.90	1.95	2.00	**2.00**	2.00	2.05	2.10	2.15	**2.15**
Fiscal balance (% of GDP)					**-5.3**					**-5.3**
Public debt (% of GDP), incl, local govt					**145**					**-**
United Kingdom										
GDP (% QoQ, ann)	1.3	0.9	-0.1	0.8	**0.6**	1.5	1.5	1.5	1.5	**1.1**
CPI headline (% YoY, avg)	3.0	2.4	3.5	3.4	**3.1**	3.2	2.9	1.9	2.1	**2.5**
BoE official bank rate (%, eop)	3.75	3.50	3.25	3.25	**3.25**	3.25	3.25	3.25	3.25	**3.25**
3-month interest rate (%, eop)	3.50	3.20	3.20	3.20	**3.20**	3.20	3.20	3.20	3.20	**3.20**
10-year interest rate (%, eop)	4.70	4.60	4.50	4.30	**4.30**	4.30	4.40	4.40	4.40	**4.40**
Fiscal balance (% of GDP)					**3.6**					**3.1**
Public sector net debt (FY, %)					**96.1**					**96.4**
EUR/USD (eop)	1.15	1.16	1.18	1.20	**1.20**	1.20	1.20	1.20	1.20	**1.20**
USD/JPY (eop)	158	158	155	153	**153**	150	148	147	145	**145**
USD/CNY (eop)	6.90	6.87	6.82	6.80	**6.80**	6.78	6.75	6.75	6.70	**6.70**
EUR/GBP (eop)	0.87	0.88	0.89	0.90	**0.90**	0.90	0.90	0.90	0.90	**0.90**
ICE Brent - US$/bbl (average)	76	91	85	77	**82**	72	70	67	63	**68**
Dutch TTF - EUR/MWh (average)	39	50	38	38	**41**	34	28	27	30	**30**

Source: ING forecasts

GDP Forecasts

QoQ% Annualised (avg)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	3.1	2.4	1.8	1.9	1.9	2.0	2.0	2.0	2.5	2.0
Japan	1.2	0.8	1.2	1.6	0.4	1.2	0.8	0.8	0.8	1.0
Germany	-0.3	1.1	1.8	2.2	1.8	2.6	2.5	2.5	0.8	2.2
France	0.6	0.4	0.4	0.6	1.2	1.4	1.6	1.2	0.8	1.0
UK	1.3	0.9	-0.1	0.8	1.5	1.5	1.5	1.5	0.6	1.1
Italy	0.3	0.4	0.9	1.3	0.9	0.6	0.8	0.6	0.7	0.8
Canada	1.8	1.5	1.4	1.6	2.2	2.0	2.1	2.0	1.1	1.9
Australia	2.3	2.2	2.3	2.2	2.5	2.5	2.5	2.5	2.3	2.5
Eurozone	0.7	0.6	1.0	1.5	1.6	1.6	1.5	1.4	0.9	1.4
Austria	1.0	0.6	1.2	1.8	1.8	1.6	1.6	1.4	0.8	1.6
Spain	1.9	1.1	2.0	1.5	2.1	2.1	2.0	2.0	2.1	1.9
Netherlands	0.9	0.9	1.5	1.6	1.4	1.5	1.1	1.0	1.4	1.4
Belgium	0.8	0.4	1.2	1.4	1.3	1.3	1.2	1.4	0.8	1.2
Greece	1.6	0.8	1.8	1.7	1.7	2.4	2.2	1.7	1.9	1.8
Portugal	1.7	1.3	1.6	1.7	2.0	2.0	2.0	2.0	2.1	1.9
Switzerland	1.2	0.6	0.8	1.2	1.2	1.6	1.6	1.2	0.5	1.2

Emerging Markets, (YoY% growth)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Bulgaria	3.1	2.8	2.7	2.5	2.5	2.5	2.5	2.7	2.8	2.5
Croatia	3.5	3.1	3.0	1.7	1.6	1.7	1.8	1.8	2.7	1.7
Czech Republic	2.5	2.8	2.7	2.8	2.8	2.8	2.7	2.7	2.7	2.7
Hungary	1.6	1.4	1.7	2.2	1.9	3.2	3.1	3.6	1.7	3.0
Poland	3.6	3.8	3.7	3.6	3.6	3.4	3.0	3.0	3.7	3.2
Romania	-0.7	-0.8	0.3	2.8	3.1	2.9	2.6	2.7	0.6	2.8
Turkey	3.2	2.8	3.6	3.8	4.4	5.5	5.2	4.9	3.4	5.0
Serbia	3.5	3.0	3.1	2.6	2.8	3.2	3.3	3.6	3.0	3.2
Azerbaijan	3.0	2.0	4.0	1.0	3.0	3.5	2.0	3.5	2.5	3.0
Kazakhstan	4.0	5.0	5.0	6.0	5.0	4.8	4.0	4.0	5.0	4.5
Russia	1.0	1.0	0.5	-0.5	0.0	0.5	1.0	1.5	0.5	0.8
Uzbekistan	6.5	6.3	6.0	7.0	6.5	5.5	6.0	6.0	6.5	6.0
Ukraine	2.1	2.6	2.5	2.8	3.2	3.4	3.5	3.5	2.5	3.4
China	4.6	4.6	4.7	4.5	4.2	4.9	4.4	4.4	4.6	4.5
India	6.5	6.6	6.8	7.0	6.5	6.5	6.5	6.5	6.7	6.5
Indonesia	4.8	4.9	5.0	5.0	5.0	5.0	5.0	5.0	4.9	5.0
Korea	2.4	2.4	1.5	2.3	1.8	1.8	1.9	1.8	2.2	1.8
Philippines	4.0	4.5	6.0	6.2	6.0	6.0	6.0	6.0	5.2	6.0
Singapore	4.5	4.0	3.0	2.0	1.8	1.8	1.8	1.8	3.4	1.8
Taiwan	10.2	7.4	6.9	3.1	4.0	4.6	5.0	5.1	6.7	4.7

Norway: Forecasts are mainland GDP
Source: ING estimates

CPI Forecasts

YoY% (avg)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	2.5	3.8	3.5	3.2	2.8	2.0	1.9	2.1	3.3	2.2
Japan	1.7	2.4	2.4	2.3	2.6	1.8	1.8	1.6	2.2	2.0
Germany	2.2	3.2	3.2	3.0	3.0	2.2	2.1	2.1	2.9	2.3
France	1.1	2.4	2.2	2.3	1.9	1.1	0.8	1.5	2.0	1.3
UK	3.0	2.4	3.5	3.4	3.2	2.9	1.9	2.1	3.1	2.5
Italy	1.5	2.3	2.6	2.6	2.2	1.7	1.6	1.9	2.3	1.8
Canada	2.1	2.8	2.9	2.5	2.1	1.7	1.8	2.0	2.6	1.9
Australia	3.7	3.7	3.2	3.2	3.0	3.0	3.0	3.0	3.5	3.0
Eurozone	2.0	3.2	3.1	2.7	2.6	1.9	1.8	2.0	2.8	2.1
Austria	2.3	2.7	2.6	2.3	2.2	1.9	1.8	2.0	2.5	2.0
Spain	2.6	3.3	2.8	2.6	2.2	2.2	2.1	2.1	2.8	2.1
Netherlands	2.3	2.5	3.0	3.1	2.6	2.3	1.7	1.6	2.7	2.0
Belgium	1.8	3.4	3.3	3.0	2.8	2.0	1.9	1.9	3.0	2.2
Greece	3.1	3.5	3.6	3.1	2.7	2.0	2.2	2.2	3.3	2.3
Portugal	2.2	2.9	2.6	2.4	2.0	2.0	1.9	1.9	2.5	2.0
Switzerland	0.4	1.1	1.0	0.7	0.6	0.6	0.8	0.8	0.8	0.7
Bulgaria	3.3	3.7	2.3	2.5	2.4	2.3	2.5	2.6	3.0	2.5
Croatia	3.8	4.1	3.3	3.2	2.4	2.6	2.7	3.1	3.6	2.7
Czech Republic	1.5	1.8	1.6	2.1	2.5	2.3	2.3	2.3	1.7	2.3
Hungary	1.9	3.4	3.6	3.9	4.0	4.2	3.8	3.4	3.2	3.8
Poland	2.5	3.6	3.7	3.3	3.1	2.2	2.0	2.3	3.2	2.4
Romania	9.7	10.4	6.0	5.4	4.2	4.1	3.9	3.7	7.9	4.0
Turkey	31.0	28.6	26.3	25.5	21.6	20.7	19.0	18.3	28.1	20.3
Serbia	2.6	3.0	2.6	3.4	3.3	3.5	3.8	4.2	3.0	3.8
Azerbaijan	5.8	5.5	4.9	4.6	4.3	4.5	4.7	4.9	5.2	4.6
Kazakhstan	11.8	11.0	10.3	10.4	10.0	9.3	8.6	8.1	10.9	9.0
Russia	5.9	5.4	4.8	5.3	4.5	4.8	5.3	4.7	5.3	4.8
Uzbekistan	7.4	7.1	7.0	7.7	8.0	7.7	7.6	7.6	7.3	7.7
Ukraine	9.2	8.5	7.5	7.0	6.5	6.3	6.1	5.7	8.1	6.2
China	0.9	1.1	1.4	1.3	1.6	1.1	1.1	1.1	1.2	1.3
India	3.0	3.7	4.1	4.3	4.5	4.8	4.8	5.0	3.8	5.0
Indonesia	3.0	2.3	2.5	2.5	2.5	2.5	3.0	3.0	2.6	2.5
Korea	2.1	3.0	2.5	2.1	1.8	1.3	1.9	2.2	2.4	1.8
Philippines	2.5	3.0	3.2	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Singapore	1.6	1.8	1.9	1.8	2.0	2.0	2.0	2.0	1.8	2.0
Taiwan	1.4	2.6	2.5	1.9	1.5	1.3	1.4	1.4	2.1	1.4

*Quarterly forecasts are quarterly average; yearly forecasts are average over the year, HICP for Eurozone economies

Source: ING estimates

Oil and Natural Gas Forecasts (avg)

	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Brent ($/bbl)	76	91	85	77	72	70	67	63	82	68
Dutch TTF (EUR/MWh)	39	50	38	38	34	28	27	30	41	30

Source: ING estimates

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Disclaimer

Assessing Europe's Competitiveness as a Location for the Life Sciences Industry

Narrative report

March 2026

Contents

Executive summary

Once a beacon of scientific excellence and a modern powerhouse in pharmaceuticals, Europe now risks further falling behind global peers, including China and the United States, which are outpacing Europe in attracting pharmaceutical investment, initiating more clinical trials and originating a greater share of novel pharmaceutical innovations.

Competitiveness has become a political and policy priority across sectors in Europe. The life sciences industry is one of Europe's most important strategic assets, delivering medical breakthroughs that are fundamental to Europe's health, economic stability, and security.

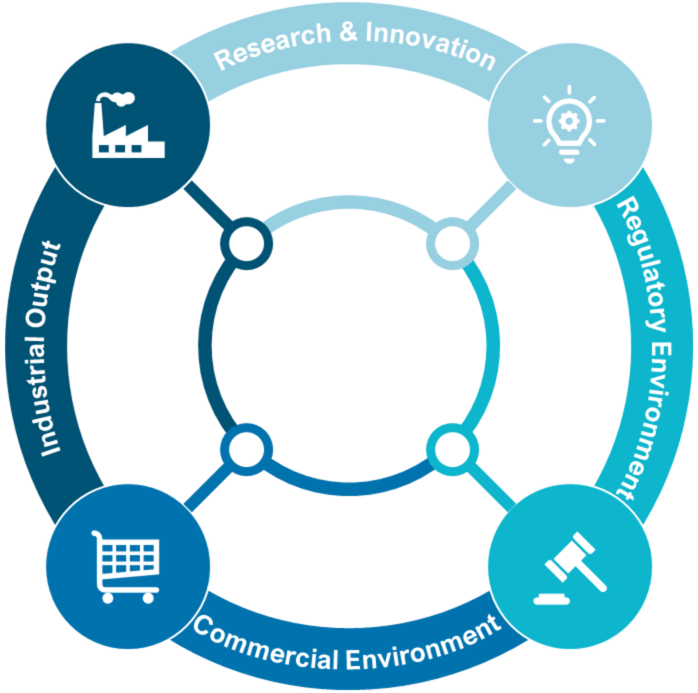
In the last two decades, Europe has experienced a decline in key drivers of competitiveness in the pharmaceutical sector. Without decisive actions to re-build a vibrant industrial ecosystem, this decline risks accelerating.

In practice, competitiveness in the life sciences sector is complex, shaped by a set of interconnected conditions. It is ultimately determined by how effectively it turns scientific capability into research and development (R&D) activities, which lead to approved products that reach patients and global markets at scale. The factors that drive life sciences competitiveness are many.

This report assesses the EU's life sciences competitiveness through this lens, focusing on four performance areas that most directly drive it: **research and innovation, the regulatory environment, the commercial environment, and industrial output.**

We compare the EU's attractiveness as a pharmaceutical investment location with that of global peers, including China, Switzerland, the United Kingdom (UK) and the United States (US).

Life sciences competitiveness is driven by four major performance areas.



Overall, the EU's strengths lie in the quality of foundational research and industrial resilience. Still, challenges in regulatory speed, IP incentives, and commercial environment risk diminishing its global competitiveness relative to peers.

Research & Innovation

- The EU performs strongly in scientific excellence (second only to the UK) and digital capabilities (though it trails the US).
- The region trails comparators in industry R&D investment growth rate, falls behind Switzerland in R&D incentives, and faces a growing gap in clinical trial attractiveness compared to China and the US. The EEA share of global trial starts declined by 50% between 2013 and 2023.
- Patent applications have grown modestly (6% over 2014–2024), but China's 170% surge highlights the EU's relative stagnation.

Regulatory Environment

- The EU is lagging in approvals of new active substances (NAS), declining 20% over the past decade (vs. China's 470% increase)
- EU's minimal use of expedited pathways (0% in 2024 vs. 53% in the US) is a pressure point.
- Regulatory approval timelines have improved (430 days in 2024, down from 464 in 2015) but remain longer than in China (390 days) and the US (356 days).

Commercial Environment

- The EU's performance in spending on pharmaceuticals is moderate at 1% of its GDP, compared to China (1.8%) and the US (2.0%)
- In terms of inward FDI, the EU performs moderately with €1.5bn in 2023, compared to China (€2.3bn) and the US (€4.3bn).
- The EU performs weakly in the launch of NAS (39% of global launches 2012–2021 vs. 85% in the US).
- Mandatory industry refunds, such as high clawback rates (up to 53% in some EU countries) compared to 0% in China, Switzerland, and the US, deter investments.

Industrial Output

- The EU excels here, with strong manufacturing investment growth (15% compound annual growth rate (2018–2022), surpassing China's 11%) and a persistent trade surplus, reflecting resilient supply chains and export capabilities. However, maintaining this position cannot be taken for granted in an increasingly competitive global landscape.

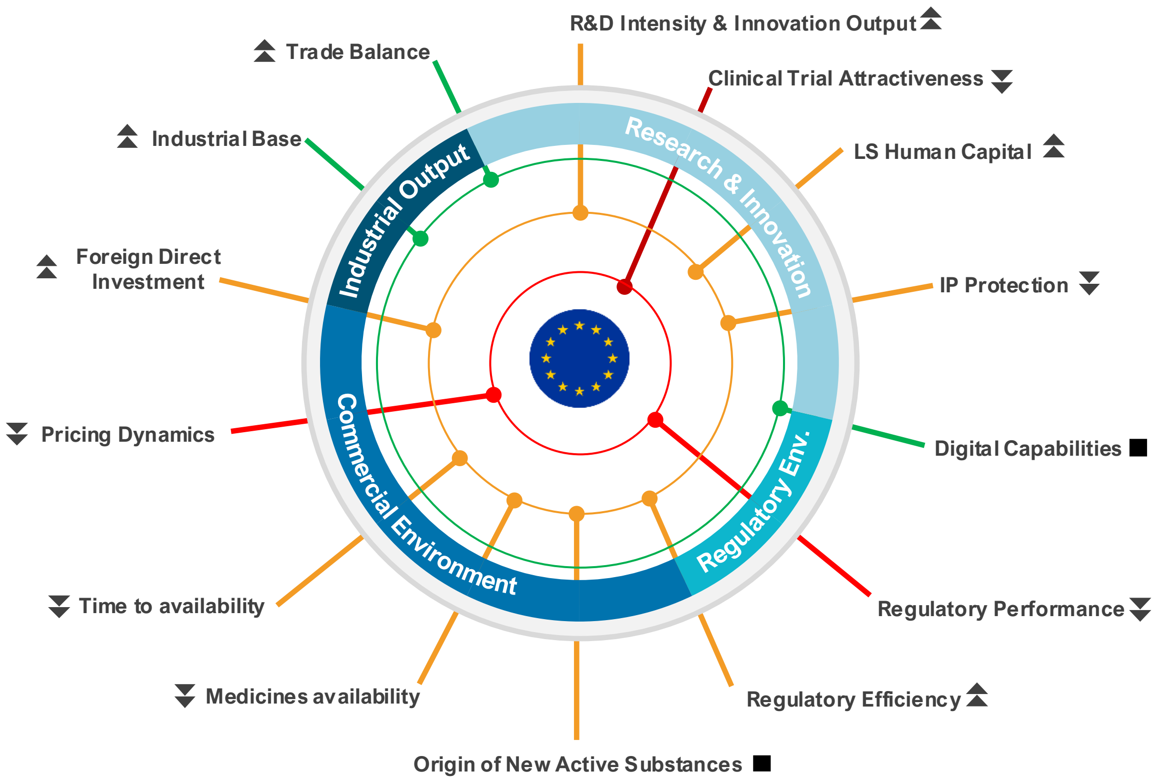
EU life sciences competitiveness at a glance

The EU retains deep scientific excellence, advanced digital capabilities, a strong industrial base, and high trade performance.

Yet, across the innovation pathway and uptake, the EU is ceding ground to the US and, increasingly, China – most visibly in the origin of new medicines, share of clinical trials, use of expedited regulatory pathways, as well as speed and breadth of medicines launches.

Europe's science and industrial base is an asset, but innovation conversion, as well as access breadth and speed, now define the EU's competitiveness gaps.

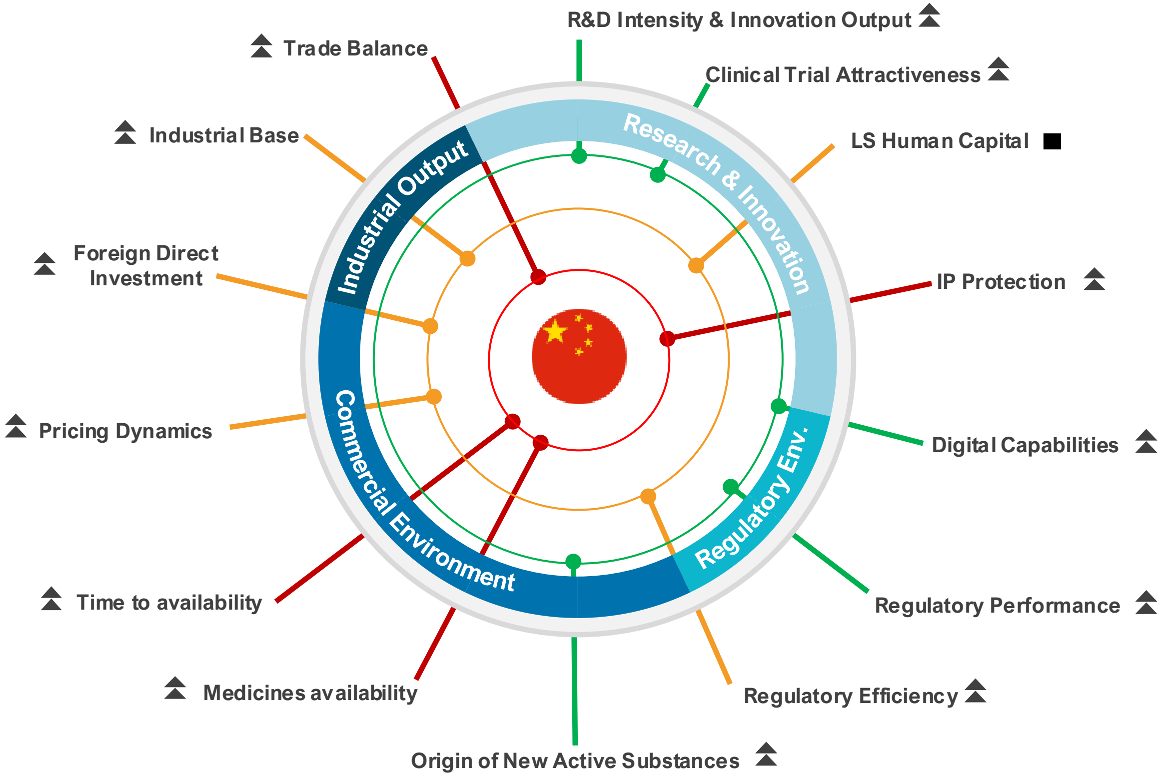
On those decisive metrics, the gap relative to the US persists, and the challenge from China is now unmistakable.

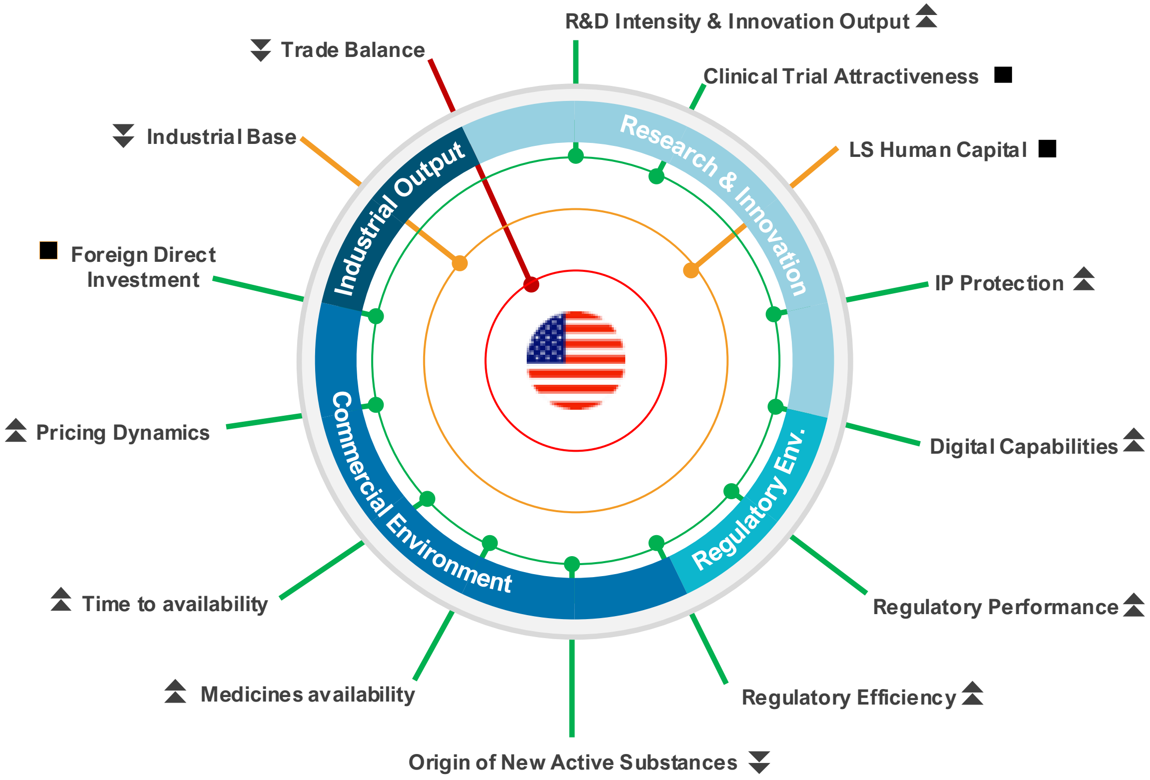


Key:

Performance (latest)	*Strong*	*Moderate*	*Lower*
Trend (5-10 years)	*Improving*	*Flat*	*Declining*

China and US life sciences competitiveness at a glance





Key:

Performance (latest)	*Strong*	*Moderate*	*Lower*
Trend (5-10 years)	*Improving*	*Flat*	*Declining*

The EU must act urgently over the next decade to strengthen its attractiveness relative to global leaders. Addressing weaknesses in key drivers of competitiveness could deliver significant gains in industry R&D investment, clinical trial activity, and the number of new active substances (NAS) originating in Europe.

Below are illustrative examples of closing the gap with global peers

Closing the gap on industry R&D investment

€105 billion

Additional industry R&D investment

- Closing the industry R&D investment gap with global peers would require the EU to attract more industry R&D investments than the current CAGR of 5.4%.
- Growing at 8.5% CAGR, a rate higher than the US's (6.4%) but lower than China's (12.1%), could yield an additional €105bn worth of industry R&D investment in the EU over a 10-year period by 2035.

Closing the gap on the share of global clinical trial starts

€17.9 billion

incremental total gross value added (GVA)

- To catch up with growth in China and North America in share of global clinical trial starts would require a 50% increase in activity compared to 2025 levels.
- This would result in:
    - Nearly €18bn in the European economy.
    - 82,000 new jobs.
    - 158,000 more patients enrolled in clinical trials.

Closing the gap on the use of expedited review pathways

200 new medicines

approved via expedited reviews

- The US is currently the leader in the use of expedited review pathways (59% of all approved NAS in 2024), many targeting areas of unmet medical need.
- Closing this gap would result in at least 20 NAS per year approved via the expedited review pathway.
- In 10 years, more than 200 NAS would have been approved using expedited review pathway in the EU – accelerating access for patients with limited or no treatment options.

Closing the gap on originating NAS

100 medicines

originating from Europe

- China currently leads in the share of NAS originated at 35% (28 NAS) in 2024, compared to 22% (18 NAS) in Europe.
- Closing the gap with China would mean an additional 10 NAS developed in Europe per year, and at least 100 NAS over a 10-year period.

Introduction

Why this report?

Competitiveness has become a political and policy priority across sectors in Europe. The life sciences sector is one of Europe's most important strategic assets, delivering innovative medicines and vaccines that are fundamental to the long-term health and security of Europe.[1,2]

Once a beacon of scientific excellence and a modern powerhouse in pharmaceuticals, Europe now risks falling further behind global peers in research and innovation, where more ambitious, dedicated strategies are driving growth and outpacing Europe in attracting talent and cutting-edge developments.[1]

Europe stands at a pivotal crossroads amid a rapidly evolving global landscape. The acceleration of breakthroughs in biotechnology, personalised medicine, and AI-driven drug discovery is reshaping human health and economies alike. This, alongside intensifying geopolitical competition, supply-chain vulnerabilities, and widening gaps in scale-up capital, has raised a direct question for policymakers: can Europe convert scientific excellence into sustained leadership in innovation, investment, and patient impact?

In practice, competitiveness in the life sciences sector is a system outcome, shaped by a set of interconnected conditions: the ability to consistently generate breakthrough innovations, translate them efficiently into clinical and commercial products, manufacture and supply at scale, attract and retain talent and capital, and deliver timely patient access.

This report assesses Europe's life sciences competitiveness through that lens – focusing not only on research output, but on the full pathway from idea to impact. We compare the EU's attractiveness as a pharmaceutical investment location with that of four comparator countries: China, Switzerland, the United Kingdom and the United States. It draws on robust quantitative and qualitative analyses and insights to highlight where Europe is leading, where it is falling behind, and where targeted reform could unlock unrealised potential and improve Europe's competitiveness in life sciences.

This publication is expected to be updated periodically to support decision-makers who must balance multiple priorities to identify key policy interventions needed to realise Europe's competitiveness ambitions for its life sciences sector.

Comparator countries included in this analysis:

- European Union (EU)
- China (CN)
- Switzerland (CH)
- United Kingdom (UK)
- United States (US)

Performance analysis:

Using publicly available information, we assessed the latest EU performance relative to global comparators, as well as performance trend over 5 -10 years, where data allow.

What factors drive life sciences competitiveness?

The factors that drive life sciences competitiveness are many. Ultimately, the competitiveness of the life sciences sector is determined by how effectively it turns scientific capability into approved products that reach patients and global markets at scale. This report frames that pathway through four performance areas that most directly drive life sciences competitiveness: **research and innovation**, the **regulatory environment**, the **commercial environment**, and **industrial output.**

Assessment framework

A set of indicators and sub-indicators under each performance area was used to assess country/regional performance

Research & innovation	Regulatory environment	Commercial environment	Industrial output
R&D intensity	*Regulatory performance*	*Origin of new active substance*	*Industrial base*
Clinical trial attractiveness	*Regulatory efficiency*	*New medicine launch*	*Trade performance*
Life sciences human capital		*Time to market (launch)*	
Intellectual property protection		*Pricing dynamics*	
Digital capabilities		*Foreign direct investment*	

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Chapter 1: Research and innovation

A conducive research ecosystem is the starting point. Countries and regions that lead in innovation are those that not only create an enabling research environment to generate new ideas but also sustain a high-throughput innovation engine that moves efficiently from proof of concept to scalable development programmes, with mechanisms and incentives that protect intellectual property.

We assessed Europe's R&D performance against comparator countries across key performance indicators:

Key performance indicators

R&D intensity
1. R&D investment growth
2. R&D investment levers
3. Scientific excellence
4. Patent applications

Clinical trial attractiveness
5. Number of global commercial clinical trial starts

Life sciences human capital
6. Life Sciences talent

Intellectual property protection
7. IP Index (IP protection strength)

Digital capabilities
8. Digital health maturity

1. Annual R&D investment growth

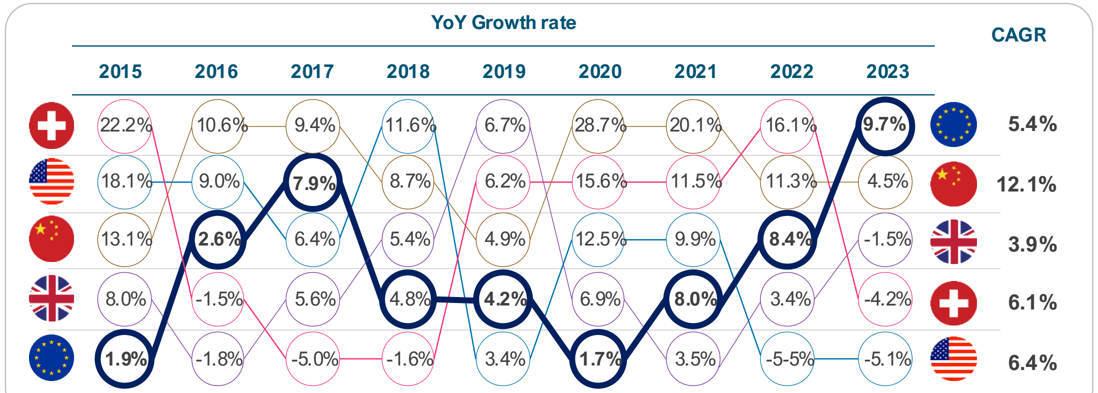
Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs weakly in growth rate of R&D investment from local and foreign companies, with growth rate about half of that in the US and two times lower than China.

Key performance indicator	Source	Latest
Growth rate (CAGR) of R&D investment by national and foreign pharma companies) [3-6]	EFPIA & ABPI	2023

1st	2nd	3rd	4th	5th
China	US	Switzerland	EU	UK
12.1%	6.4%	6.1%	5.4%^	3.9%

^EFPIA countries aggregate (excluding Switzerland and the UK)



Acronyms: CAGR - compound annual growth rate

NOTE: *UK growth rates were derived from ABPI analysis of ONS 'business expenditure on research and development UK: 2024' available here. Data prior to 2022 is based on estimates following a methodology change in how UK BERD data is collected.*

Over the last decade, European pharmaceutical R&D investment has grown steadily, but at a pace that may not fully match global shifts.[3-7] Annual growth rates in R&D investment by national and foreign pharmaceutical companies from 2015 to 2024 reveal distinct regional trends, as measured by the CAGR.

In EU countries, the CAGR is 5.4%, indicating steady growth. However, this pace falls behind growth rates in comparators. China exhibits the highest CAGR of 12.1%, followed by the US (6.4%) and Switzerland (6.1%). The lower growth rates in industry R&D investments in the EU highlight vulnerabilities in the EU's pharma ecosystem to attract investments.

The rapid acceleration observed in China, where investments surged notably in 2019 (28.7%) and 2020 (20.1%), reflects aggressive policy support for innovation and biotechnology. In absolute terms, the US is the global leader. R&D investments rose 65%, from €43bn in 2015 to €71bn in 2023, maintaining dominance, with expenditures nearly double those of the EU27.[3-6]

These trends have profound implications for the EU's pharma sector. Lagging growth relative to the US and China risks diminishing Europe's innovation leadership, potentially resulting in fewer breakthroughs in areas such as oncology and rare diseases.

While the EU's pharmaceutical R&D has grown reliably, sustaining global competitiveness cannot be taken for granted. In 2024, R&D investment growth in the EU was driven mainly by the health sector.[7] This demonstrates the sector's importance and the need for targeted policy measures to strengthen Europe's global standing and competitiveness.

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1. Annual R&D investment growth

Strong EU performance | Moderate EU performance | Lower EU performance

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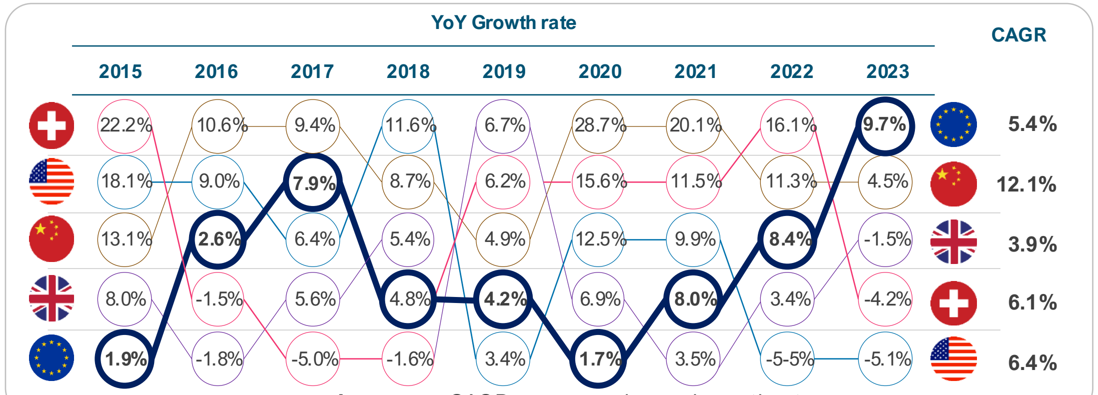
12.1%	6.4%	6.1%	5.4%^	3.9%

^EFPIA countries aggregate (excluding Switzerland and the UK)



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The rapid acceleration observed in China, where investments surged notably in 2019 (28.7%) and 2020 (20.1%), reflects aggressive policy support for innovation and biotechnology. In absolute terms, the US is the global leader. R&D investments rose 65%, from €43bn in 2015 to €71bn in 2023, maintaining dominance, with expenditures nearly double those of the EU27.[3-6]

These trends have profound implications for the EU's pharma sector. Lagging growth relative to the US and China risks diminishing Europe's innovation leadership, potentially resulting in fewer breakthroughs in areas such as oncology and rare diseases.

While the EU's pharmaceutical R&D has grown reliably, sustaining global competitiveness cannot be taken for granted. In 2024, R&D investment growth in the EU was driven mainly by the health sector.[7] This demonstrates the sector's importance and the need for targeted policy measures to strengthen Europe's global standing and competitiveness.

2. R&D investment levers

Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs moderately in the number and value of R&D investment levers. The EU lags Switzerland and the UK in the number of measures, and China and Switzerland in terms of their value.

Key performance indicator	Source	Latest	
Implied subsidy rate on R&D expenditure (range) [8-12]*	EY & other sources	2023	

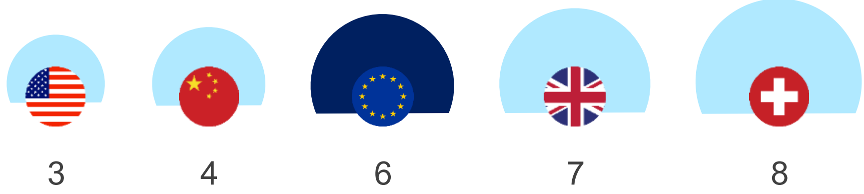
1st	*2nd*	*3rd*	*4th*	*5th*
22% - 32%	13% - 22%^	12% - 27%	1% - 7%	0% - 1%

**Simple benchmark comparison (large profitable firms). ^EU27 average*

***NOTE:** Implied or assumed subsidy rate can vary within a country/region. Switzerland has historically not operated a federal R&D tax credit. R&D expenses are simply deductible as normal business costs, resulting in a neutral tax treatment. Switzerland instead incentivizes commercialisation of IP, not the inputs to R&D*

There are 16 different types of R&D investment levers across countries [6]

Number of R&D investment levers available in comparator countries



R&D investment levers, such as tax credits and cash grants, are a core lever of competitiveness because they directly affect whether companies invest in long-term, high-risk research and where that investment is located.[12]

Based on the breadth of R&D support measures available, our analysis shows that the EU countries exhibit a relatively moderate level of availability, with an average of 6 instrument types per member state. [13] However, there is significant variation across the EU. This ranges from high performers such as the Netherlands (10) and Belgium (9) to lower performers like Romania (3).[13] Such diversity highlights the fragmented nature of R&D support in the EU. The EU's average (6) surpasses that of China (4) and the US (3), but trails Switzerland (8) and the UK (7).[13]

R&D tax credits or deductions are among the most commonly used investment levers across sectors as they play a decisive role in investment decisions. The value of R&D tax incentives (i.e., implied subsidy rate on R&D expenditure) widely varies across (and even within) countries. Data from OECD INNOTAX show that implied subsidy rate on R&D expenditures across the EU ranges from 0% to 39%.[20] R&D tax deductions typically range from 20% to 36% in high-signal EU member states, including France [14], Germany [15], and the Netherlands [16] In Switzerland, the R&D super-deduction (cantonal option) could be up to 50% [17], and up to 200% in China [18], making China and Switzerland more attractive for investment.

The EU, with an average implied subsidy rate of 16%, falls behind China (32%).[9-12,19,20,25] This, together with weakening EU incentive frameworks, particularly the reduction of the baseline RDP, may deter long-term investment. [27]

3. Scientific excellence

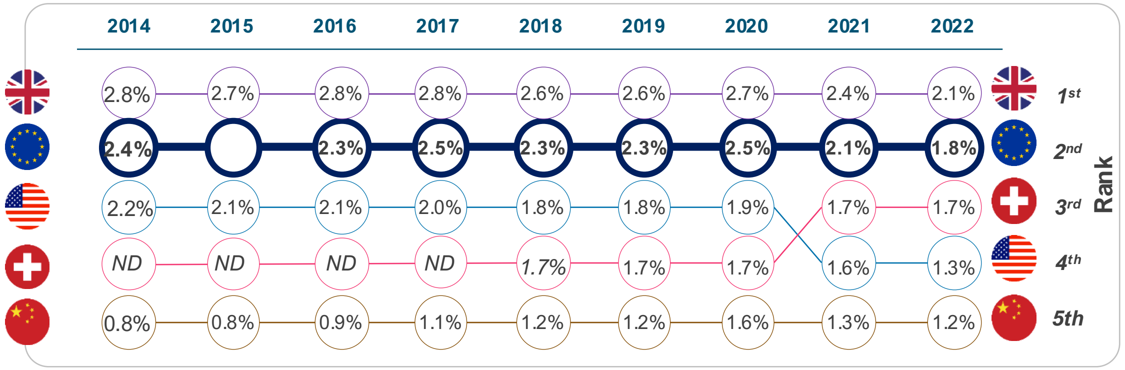
Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs strongly in the percentage of medical sciences publications which are amongst the most highly cited (top 1%) globally. Over the last decade, the EU has ranked only second to the UK on this performance indicator.

Key performance indicator	Source	Latest
Highly cited papers published (% of total) [28-30]*	Multiple sources	2024

1st	2nd	3rd	4th	5th
2.1%	1.8%^	1.7%	1.3%	1.2%

*Percentage of each country's medical sciences publications that are amongst the top 1% highly cited.
^Average of France, Germany, and Italy



Acronym: *ND – no data*

The proportion of a country's (or region's) medical sciences publications that rank among the most highly cited globally (top 1%) serves as a key bibliometric indicator of research quality, scientific excellence, and impact in the health sector.

This indicator is highly relevant to pharmaceutical competitiveness, as it underscores the strength of a country's foundational research ecosystem, which is essential for driving pharmaceutical innovation.

Relative to comparator countries, Europe's performance indicates resilience in research excellence, positioning it favourably against the UK and Switzerland—both renowned for high-impact outputs—and ahead of China and the US in terms of publication quality per output. [28-30]

Nonetheless, the trend over the last decade reveals a concerning reality: a 23% decline in performance in Europe between 2014 and 2022.

Although China lags comparators on this indicator, an interesting development is that it has increased by almost 50%, rising from 0.8% in 2014 to 1.2% in 2022. [30]

This demonstrates the scale of investment that drives China's transition toward scientific excellence and research impact.

If this trend continues, China is likely to overtake Europe soon, further eroding European competitiveness in research impact and scientific excellence.

4. Patent applications

Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs moderately in terms of the number of patent applications relative to comparator countries. The share of global PCT applications from Europe has declined over the last decade.

Key performance indicator	Source	Latest
Patent (PCT) applications per capita [31]*	WIPO	2024

Rank	1st	2nd	3rd	4th	5th
Per capita	60.3	15.9	10.2	8.5	4.9
Total number	5,324	54,087	45,711	5,861	70,160

**This is the number of PCT applications per capita (across all sectors). Population data retrieved from the World Bank*

	2014	2017	2018	2019	2020	2021	2022	2023	2024	CAGR
	50.2	53.1	53.6	53.7	56.5	61.9	61.0	60.7	60.3	+2%
	19.3	17.4	17.2	17.6	17.9	17.9	17.7	16.4	15.9	-2%
	9.5	10.5	10.7	10.6	10.7	10.5	10.1	10.2	10.2	-1%
	8.2	8.4	8.5	8.6	8.8	8.6	8.4	8.1	8.5	+0.4%
	1.8	3.5	3.8	4.1	4.8	4.9	4.9	4.9	4.9	+10%

CAGR – compound annual growth rate

Patent activity is a critical driver of competitiveness, as it reflects a country's ability to generate, protect, and commercialise new technologies, and to attract investment in high-value, knowledge-intensive sectors.

The number of Patent Cooperation Treaty (PCT) applications filed by entities within a country or region, therefore, serves as a widely used indicator of innovation performance (not specific to pharmaceutical or biotechnology sectors). Analysis of annual PCT filings from 2014 to 2024 indicates a **slower trajectory for EU countries** compared with global peers. [32]

In absolute terms, China (70,160) ranked first in 2024. China (58,990) overtook both the US (57,840) and the EU (54,040) in 2019. On a per capita basis, Switzerland leads, with an average of 56 applications per 100,000 population per year, compared with 18 in the US, 10 in Europe, and 4 in China.[17] Over the period, the EU recorded a CAGR of -1% per year, below China (10% per year) and Switzerland (2% per year), showing a relative decline.[31]

Sector-specific data shows a similar pattern in pharmaceuticals. According to the European Patent Office (EPO), pharmaceutical patent filings declined by 13% in 2024, even as total European applications increased slightly by 0.3%. [32]

If sustained, this trend could gradually weaken the EU's pharmaceutical competitiveness and long-term innovation capacity.

5. Clinical trial attractiveness

Strong EU performance

Moderate EU performance

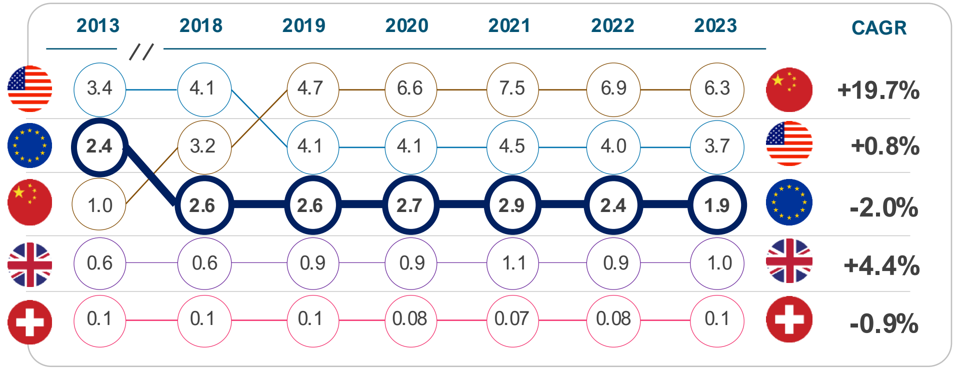
Lower EU performance

Europe is losing ground in attracting clinical trials compared to its global peers. The EU's share of global industry clinical trial starts declined by 50% between 2013 and 2023.

Key performance indicator	Source	Latest
Number of global clinical trial starts [33]	IQVIA	2023

1st	2nd	3rd	4th	5th
			*	*
6,392	3,747	1,978	1,025	101

**This represents the number of commercial clinical trial starts only*
US data represent the North America region. EU data represent the EEA region



Number of clinical trial starts in '000 [33]

Clinical trial attractiveness refers to the extent to which a country or region is appealing to pharmaceutical sponsors for conducting clinical studies. This attractiveness is crucial for a country's competitiveness in the pharma sector, as it directly influences innovation, economic growth, and the country's global position within the industry.[3,33-35]

In the last decade, there has been a major shift in the geographic distribution of trials. In 2013, North America (including the US), the European Economic Area (EEA), and the rest of Europe (including Switzerland and the UK) accounted for 53% of global clinical trial starts (commercial and non-commercial). As of 2023, this figure stood at 33%. Over the same period, China's share increased from 8% to 29%, representing approximately 20% annual growth.[34,35]

Similarly, analysis of global commercial trial starts shows a similar trend. While EEA countries (except Spain) are losing share of global commercial trial starts, declining from 22% in 2013 to 12% in 2023, China continues to gain share, growing from 5% to 18% in that period.[34]

The rapid growth seen in China is not entirely surprising given its modernising society and market size. It is the worsening loss of market share in the EU that is concerning. Structural challenges, such as fragmented ecosystem and longer clinical trial start-up timelines have prevented the EU from growing to the level of opportunity that exists within the EU itself.[34] To strengthen its position as a global innovation leader, the EU should prioritise a more consistent implementation of the EU Clinical Trials Regulation and streamline processes for conducting multi-country trials.

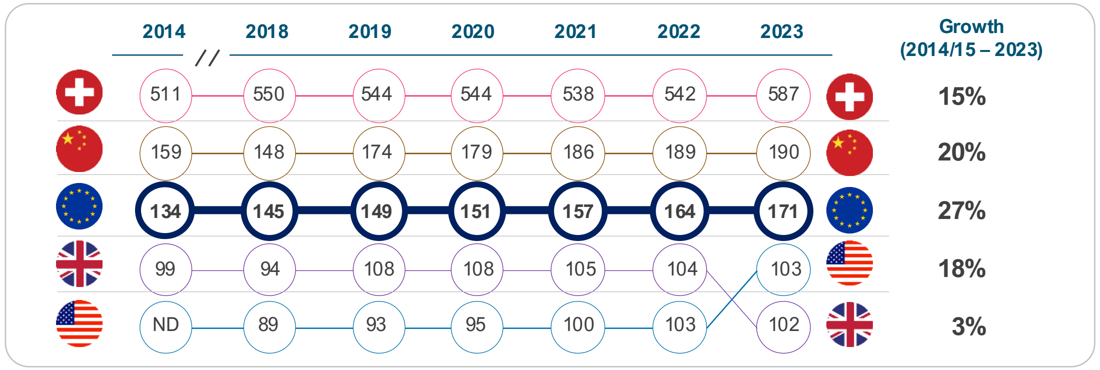
6. Life sciences talent

Strong EU performance | Moderate EU performance | Lower EU performance

Compared with comparator countries, EU countries perform moderately in human capital in the life sciences, as measured by the number of employees per capita in the pharmaceutical sector.

Key performance indicator	Source	Latest
Number of life sciences employees per capita (100,000 population) [3-7,36-37]	Multiple sources	2023

1st	2nd	3rd	4th	5th
587	190	171^	103	102



Percentage of graduates with statistics, maths, or natural sciences degrees (2023) [28]:

8.3	7.9	6.3	5.2^

^Aggregate of EU27 countries. ND – No data

Stock of human capital is an important driver of competitiveness in any sector.[38] A robust employment base fosters innovation by concentrating skilled professionals, such as scientists, engineers, and clinicians, who drive advancements in drug discovery, development, and manufacturing. Countries with high employment density in the life sciences sector are likely to attract higher levels of foreign investment if other competitiveness levers are in place.[38,39]

The EU27 demonstrated strong growth of 27%, rising from 134 to 171 employees per 100,000 population. This performance reflects collective advances across the EU in research, scientific excellence, and innovation, aimed at enhancing the sector's attractiveness.

In comparison, China achieved notable growth of 20%, elevating from 159 to 190, driven by substantial investments in biotechnology and pharmaceutical infrastructure amid its economic expansion.

Switzerland, already a leader in density, grew by 15% from 511 to 587, maintaining its position as a global hub for pharmaceutical giants and high-value research and development. The country's dynamism has been partly attributed to a powerful combination of world-class academic research, strong commitment to R&D, and a symbiotic relationship between academia and industry.[39]

Lastly, in terms of talent pipeline (graduates with natural sciences degrees), the EU falls behind comparators, with 5.2% compared to the US (8.3%).[40]

7. Intellectual property protection

Strong EU performance | Moderate EU performance | Lower EU performance

Relative to comparators, the EU is a moderate performer in intellectual property protection, only better than China.

Key performance indicator	Source	Latest
IP Index (IP protection strength) [41]	USCC	2024

1st	2nd	3rd	4th	5th
95%	94%	86%	82%^	58%

^EU average (unweighted)

	2014	2017	2018	2019	2020	2021	2022	2023	2024	Growth
	95%	93%	95%	95%	95%	95%	95%	95%	95%	0%
	92%	93%	95%	94%	94%	94%	94%	94%	94%	2%
	ND	85%	88%	88%	85%	86%	86%	86%	86%	-5%
	91%	81%	84%	83%	85%	84%	85%	85%	82%	-10%
	39%	41%	48%	48%	51%	53%	56%	58%	58%	49%

ND – no data

The U.S. Chamber of Commerce's International Intellectual Property (IP) Index provides a comprehensive annual evaluation of IP frameworks across global economies, assessing factors such as patent rights, copyright protections, enforcement mechanisms, and adherence to treaties.[41]

The EU demonstrates a robust IP framework, consistently achieving average scores in the 73–88% range from 2014 to 2024, but performs below its peers, falling behind the US, UK, and Switzerland.[41]

IP systems in the US, UK, and Switzerland are generally characterised by stronger enforcement, fewer limitations on patentability, and more comprehensive incentives to support innovation.[41]

In the pharmaceutical sector, the EU's IP competitiveness has remained relatively stable. However, in recent years, it has shown signs of decline (-10%), partly owing to harmonisation reforms that introduce uncertainties in regulatory data protection and supplementary protection certificates.[41,42]

While China lags other comparator countries in IP protection strength, its upward trajectory (growing 49% over the last decade) signals China's growing emphasis on IP as part of broader industrial strategies, potentially narrowing the gap with global peers if current trends continue.[27,41-43]

In sum, the EU's pharmaceutical IP regime is less competitive than many of its peers and is already signalling a decline. Maintaining robust IP protections is essential to strengthening its competitiveness.

8. Digital health maturity

Strong EU performance | **Moderate EU performance** | **Lower EU performance**

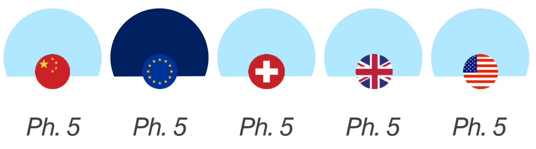
Similar to comparator countries, the EU is a strong performer in digital health maturity. The available dataset for this indicator is immature and warrants monitoring in the future.

Key performance indicator	Source	Latest	EU
Digital maturity index [44]	WHO	2023	

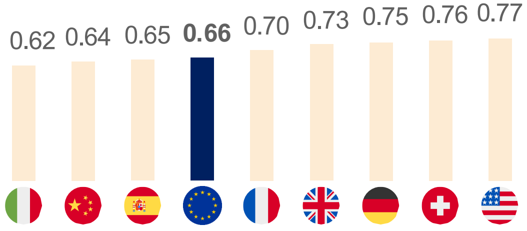
Most EU countries and the four comparator countries are ranked at being at the highest phase of digital health maturity by the Global Digital Health Monitor in 2023

The highest level of maturity is Phase 5, which signifies fully implemented, optimised, sustainable, and institutionalised systems that encompass advanced data governance structures and security, privacy, and device regulation, as well as robust infrastructure and scaled digital services.

Digital health maturity (GDHM 2023) [44]



AI Preparedness Index – AI Infrastructure (IMF 2025) [45]



Digital health, including data and artificial intelligence (AI) capabilities, continues to revolutionise healthcare delivery. Pharmaceutical companies are leveraging digital tools and AI capabilities to enhance efficiency in R&D execution and business planning. This indicates that countries with these capabilities are likely to increasingly become competitive.

The Global Digital Health Monitor (GDHM) provides a recent assessment of digital health maturity across a set of pillars not limited to digital/data governance, investment, and infrastructure, among other indicators. In this index, countries are rated on a scale from Phase 1 to 5 for each pillar, based on their level of maturity. The latest index (2023) rated most EU countries (18 of 27) and comparator countries at Phase 5 maturity.[44]

In terms of AI infrastructure, the AI Preparedness Index (2025) developed by the International Monetary Fund ranks EU countries and comparators highly in AI capabilities, with the US (0.77) leading the way, ranking above other countries, such as Switzerland (0.76), the UK (0.73), China (0.64) and many EU countries, including Germany (0.75) and France (0.70).[45] The average across the EU is 0.66, much lower than in the US, indicating some competitive advantage for the US. Reports from countries such as Germany suggest a **lag in the digitalisation of pharmaceutical R&D** relative to the US. [46]

Other assessments of AI capabilities, such as AI commercialisation [47] and AI research ecosystem [48], show similar rankings, with the US on top. The US demonstrates advanced digital capabilities driven by an ecosystem in which digital and AI tools are deeply integrated into research and patient care.[49]

Chapter 2: Regulatory environment

The regulatory environment shapes both speed to market and investment confidence. Competitive systems combine rigorous standards with clarity, consistency, and modernised pathways suited to assess increasingly complex health technologies. A predictable, efficient, and innovation-friendly regulatory system enhances a country's competitiveness.

We assessed Europe's regulatory performance and efficiency against comparator countries across key performance indicators:

Key performance indicators

Regulatory performance

9 Number of new active substances (NAS) regulatory approvals

10 Proportion of NAS approved via expedited review pathways

Regulatory efficiency

11 Median regulatory approval timeline for NAS (days)

9. Regulatory performance – standard pathway

Strong EU performance

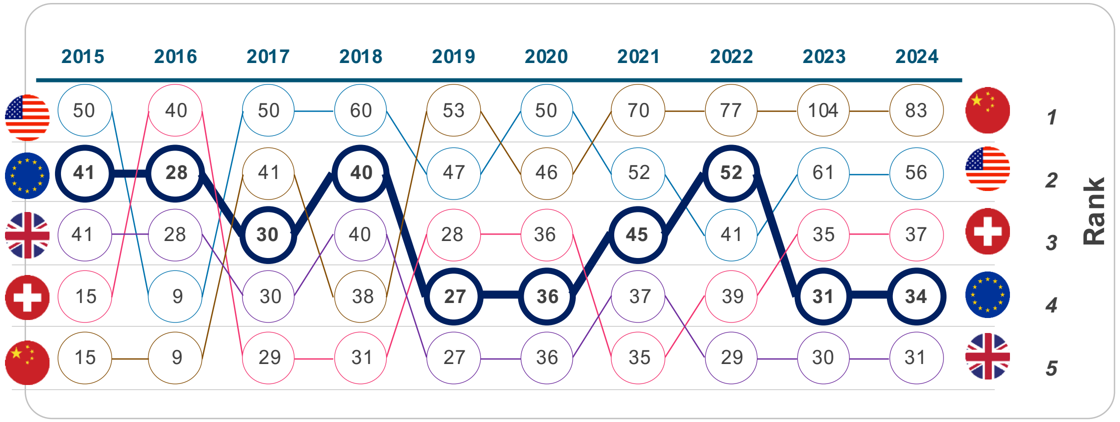
Moderate EU performance

Lower EU performance

EU regulatory output has grown more slowly than key global comparators over the past decade. Over the same period, China's performance increased markedly, driven by a sharp rise in NAS approvals.

Key performance indicator	Source	Latest
Number of new active substances (NAS) regulatory approvals [50-57]	Multiple sources	2024

1st	*2nd*	*3rd*	*4th*	*5th*
83	56	37	34	31



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020).

The regulatory environment plays a pivotal role in determining a country's pharmaceutical competitiveness.

Jurisdictions with streamlined regulatory approval systems that facilitate timely market entry for pharmaceutical innovations, which in turn accelerates patient access and reduces development costs, are likely to attract greater private investments, thereby enhancing competitiveness.

Number of new active substance (NAS) approvals is a good indicator of regulatory performance. Analysis of NAS approvals between 2015 and 2024 indicates that the EU's regulatory output has grown more slowly than that of some global peers over the past decade. Over this period, the EU recorded a relative decline of around 20% in NAS approvals compared with global comparators. [50-52]

Over the same period, China's performance improved by over 470%, from 15 approvals in 2015 to 83 in 2024, with a peak of 104 in 2023. [50-52, 56-61]

These trends do not reflect a lack of regulatory quality in the EU, but rather point to differences in system scale, resourcing, and the adoption of bold regulatory reforms in China.

As global R&D activity continues to expand, ensuring that the EU regulatory framework is adequately resourced and continuously modernised will be essential to support timely assessments, uphold high standards, and sustain the EU's role in global health standard-setting.

10. Regulatory performance – expedited reviews

Strong EU performance

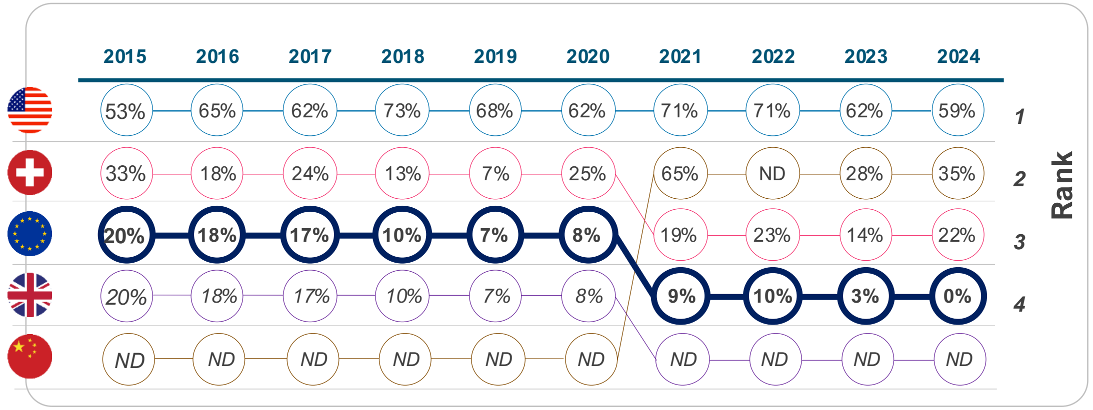
Moderate EU performance

Lower EU performance

Use of expedited review pathways differs across regions, with low uptake in the EU compared with global peers.

Key performance indicator	Source	Latest	
Proportion of NAS approved via expedited review pathways [50-52, 56-57]	Multiple sources	2024	

1st	2nd	3rd	4th	5th
59%	35%	22%	0%	*ND*



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020). ND – no data

Expedited review pathways refer to 'Accelerated Assessment' by EMA (EU), 'Fast Track' by Swissmedic (Switzerland), and 'Priority Review' by the FDA/NMPA (US/China). These pathways are designed to expedite the evaluation and approval of pharmaceutical innovations.

Empirical evidence demonstrates that the EU employs expedited regulatory pathways significantly less frequently than global peers.[50-52,58-62]. In the last decade, the utilisation of expedited review mechanisms for regulatory approval of NAS within the EU declined from 20% in 2015 to 0% in 2024. By comparison, in 2024, expedited pathways accounted for 35% of approvals in China and 53% in the US.[50-52,58-62]

China's recent regulatory reforms have played a crucial role in creating a robust end-to-end regulatory framework that aligns medicine approval processes with global standards.[61,63] In the US, expedited regulatory pathway tools are better interconnected; i.e., once a medicine receives Breakthrough Designation, it often receives Priority Review, Rolling Review, and Accelerated Assessment. In addition, the US has recently launched the FDA Commissioner's National Priority Voucher to further expedite the evaluation of new medicines aligned with critical US national health priorities.[64]

Whereas in the EU, expedited review is a complicated process. Sponsors must first apply for Priority Medicines Scheme designation, then reapply separately to receive Accelerated Assessment, and reapply again to receive Conditional Marketing Authorisation.[65] Simplifying procedures, while maintaining high regulatory standards, would strengthen the EU's overall regulatory performance.

ND – no data

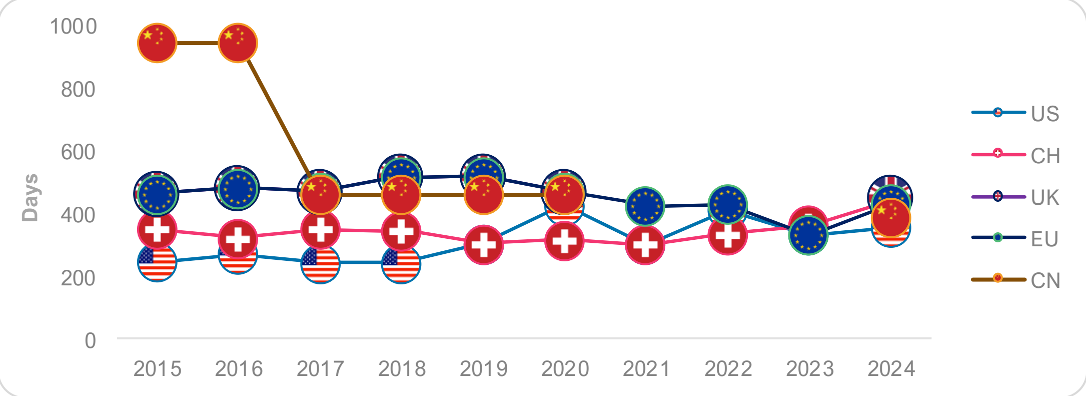
11. Regulatory efficiency – approval timeline

Strong EU performance | Moderate EU performance | Lower EU performance

Regulatory speed has improved in the EU over the last decade, performing moderately relative to peers in recent years.

Key performance indicator	Source	Latest	EU
Median regulatory approval timeline for NAS (days) [50-52,66]	CIRS; ABPI	2024	Moderate

1st	2nd	3rd	4th	5th
US	CN	EU	CH	UK
356	390	430	440	450*



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020).

Regulatory speed is a key driver of competitiveness, as pharmaceutical sponsors are more likely to allocate resources to jurisdictions with predictable, expedited regulatory timelines.

Across jurisdictions, regulatory speed can be assessed using median approval timelines (from regulatory submission to regulatory approval).

Over the last decade, the EU's median approval timelines in the standard pathway declined, from 464 days in 2015 to 430 days in 2024, with a notable low of 333 days in 2023. [50-52]

However, the headline time-to-approval remains longer than that of China and the US. The median difference between China and the EU was ~40 days (430 vs 390 days), and ~74 days (430 vs 356) for the US vs the EU. Historically, the gap between the US and the EU has often been larger. [50-52]

While these trends suggest improvement in the EU's regulatory efficiency, the EU still lags behind global peers. The European Commission has proposed reducing standard assessment timelines to 180 days. This target is not yet a legal requirement until the legislation for the new EU Pharmaceutical package is formally adopted.[50-52]

Meeting this target will help narrow competitiveness and improve the EU's attractiveness to investors.

The commercial environment determines the extent to which innovation is funded and available to patients. Where commercial conditions are strong, companies often scale locally rather than relocating, and health systems adopt innovations without delay.

We assessed Europe's commercial environment performance against comparator countries across key performance indicators:

Key performance indicators

Origin of new active substances
- 12 Origin of regulatory-approved NAS (% of all NAS)

Medicine launch
- 13 Launch of new active substances (NAS) (% of all NAS)

Time to market (launch)
- 14 Median time from global launch to local launch (months)

Pricing dynamics
- 15 Cost-effectiveness thresholds for pricing and reimbursement
- 16 Pharmaceutical expenditure as a share of GDP
- 17 Mandatory industry refund rate on pharmaceutical revenues (%)

Foreign direct investment
- 18 Inward FDI in pharma (EUR)

12. Origin of new medicines

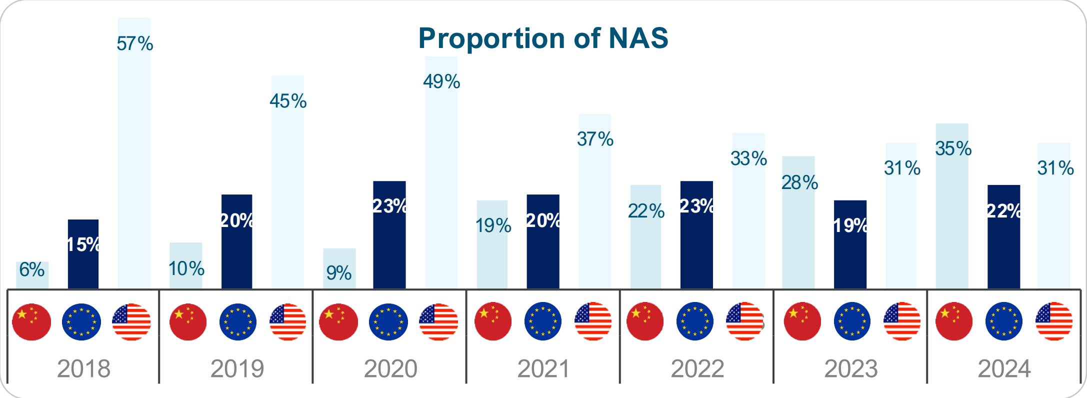
Strong EU performance | Moderate EU performance | Lower EU performance

Europe's performance in originating new active substances has remained broadly stable, while growth has accelerated in China and slowed in the United States.

Key performance indicator	Source	Latest	
Origin of regulatory-approved NAS (% of all NAS) [3-6]	EFPIA	2024	Moderate

1st	2nd	3rd
China	**United States**	**Europe***
35%	31%	22%

**This includes NAS originating from Switzerland and the UK as well.*



In the early 2000s, Europe accounted for the largest proportion of globally originating NAS. Over time, the distribution has evolved, with China increasing its contribution and a more mixed trend in the US.

Europe's performance (including Switzerland and the UK) has remained relatively stagnant from 2021 to 2024, fluctuating between 17 and 19 NAS annually. Data show that EU-headquartered firms launched only ~130 NAS in 2015–24, compared with ~257 by US firms, a stark difference in innovative output. [3-6] Europe's trajectory differs from that of the US, which has continued to originate a high number of NAS, averaging 24 to 35 per year. This performance is associated with its large, unified market and supportive ecosystem – including strong venture capital funding, a favourable reimbursement environment, fast regulatory pathways, and substantial public and private R&D investment.[67]

China has also recorded rapid growth in NAS origination in recent years, overtaking Europe in 2023 and exceeding both Europe and the US in 2024. Output increased from 4 NAS in 2018 to 28 in 2024. Over the period 2018–2024, China's growth rate exceeded that observed in Europe, reflecting sustained investment in research infrastructure, government support measures, improvements in IP protection, and the expansion of domestic biopharmaceutical capabilities. [3-6, 67,68] Furthermore, NASs originating from China are increasingly being launched in both the US and Europe.[56]

While Europe continues to originate a stable number of new medicines, strengthening the conditions that support innovation will be important to maintain its role in an increasingly competitive global landscape.

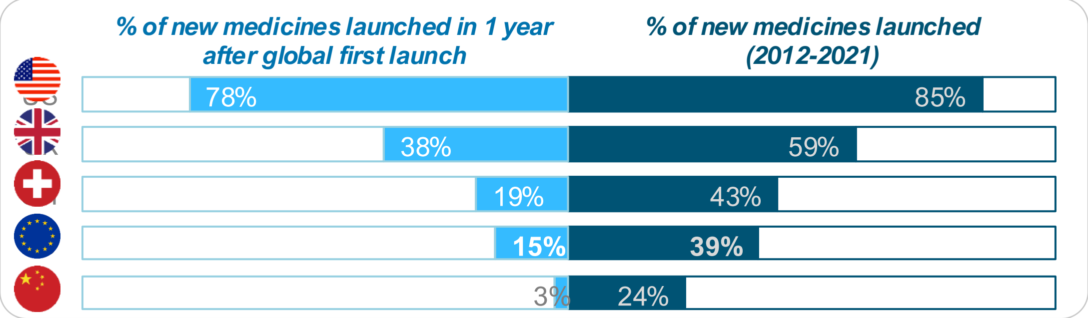
13. Medicine launch

Strong EU performance | Moderate EU performance | Lower EU performance

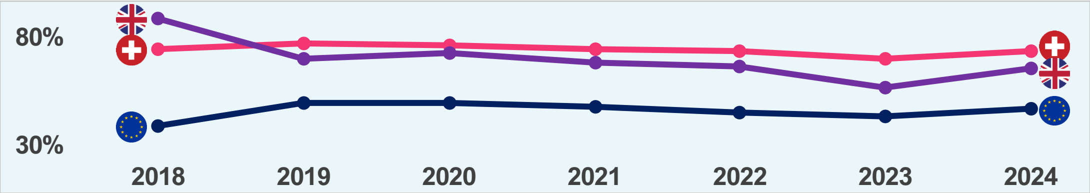
The EU's performance is moderate in terms of the launch of new medicines, compared to the US and the UK.

Key performance indicator	Source	Latest	
Launch of new active substances (NAS) (% of all NAS) [69]	PhRMA	2021	

1st	2nd	3rd	4th	5th
85%	59%	43%	39%	24%



Rate of availability (%) of NAS in Europe (EFPIA W.A.I.T.) [70]



The launch of new medicines (i.e., new active substances) is a core indicator of pharmaceutical competitiveness because it captures how effectively a health system converts scientific innovation into commercial reality, patient access, and sustained investment.

An analysis of data on 460 new medicines launched globally between 2012 and 2021 shows that the EU lags all comparator countries except China.[69]

On average, EU countries achieved a launch rate of 39%, substantially lower than the US of 85%. When focusing on timeliness, the EU average for launches within one year of global first launch is 15%, compared with 78% in the US, 38% in the UK, and 19% in Switzerland.[69]

Disparities persist even among leading EU performers, such as Germany (61% launched overall and 44% within one year), compared to France (52% launched overall and 23% within one year). While Germany tops Europe, it remains well below US levels.[69] The EU's relative underperformance in launch speed jeopardises its competitiveness, capturing only 16% of NAS sales (2019 – 2023) in top markets compared to 67% in the US.[70,71]

A similar trend is observed in NAS availability (public reimbursement) **with wide variability within the EU**. In 2023, an average of 46% of NAS was reimbursed in the EU, lower than in Switzerland (73%) and the UK (65%).[70]

The EU's lagging performance in NAS launches and availability poses significant risks to pharmaceutical competitiveness, diminishing its role as a global innovation leader.[71]

14. Time to market (launch)

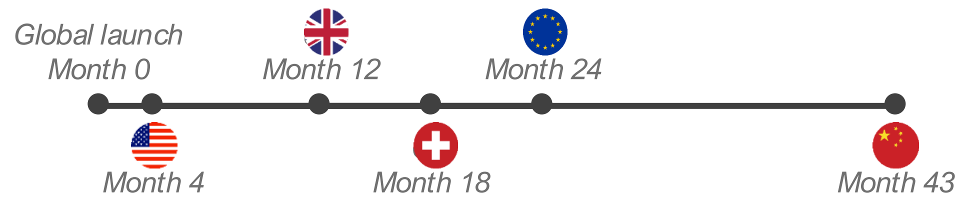
Strong EU performance | Moderate EU performance | Lower EU performance

Compared with peers, the EU performs moderately on the time to local launch of new medicines (NAS).

Key performance indicator	Source	Latest
Median time from global launch to local launch (months) [69]	PhRMA	2023

1st	2nd	3rd	4th	5th

Average months from global first launch to local launch



The EU's performance on the time-to-launch of new medicines, measured as the average number of months between a new medicine's global first launch and its local launch, reveals notable disparities relative to global peers. The average delay in the EU stands at 24 months, though it widely varies across EU countries, **ranging from 128 days (4 months) in Germany to 840 days (28 months) in Portugal**. A similar trend is highlighted in EFPIA's W.A.I.T Indicator Survey.[70]

In comparison, the US demonstrates the shortest average delay at 4 months, enabling significantly faster access for patients. The UK and Switzerland follow with average of 12 and 18 months, respectively – both outperforming the broader European average. [69] The US consistently demonstrates superior performance in both the proportion of NAS made available and the speed of their launch, underscoring a regulatory and market environment conducive to rapid innovation deployment. The wide gap with the US underscores challenges to the EU's pharmaceutical competitiveness. Prolonged delays in the launch of new medicines may deter global investors from prioritising EU markets for initial launches, as seen in the US's dominance in global first introductions.

Fragmentation and slower access, with variations among member states, signal an unattractive environment. Such patterns disincentivise R&D investment within the EU, limit patient access, and erode the region's role as a hub for pharmaceutical innovation.[71]

To enhance competitiveness, the EU would benefit from policies that streamline and expedite regulatory approvals, strengthen IP protections, and promote reimbursement systems that truly reward innovation, aligning more closely with leading performers such as the US.

15. Cost-effectiveness thresholds

Strong EU performance | Moderate EU performance | Lower EU performance

The EU shows moderate performance in willingness to pay (WTP) for a standard unit of health benefit (QALY) gained from new medicines at an average cost per QALY of €43,500.

Key performance indicator	Source	Latest
Cost-effectiveness thresholds for pricing and reimbursement [72-75]	Multiple sources	2024

1st	2nd	3rd
High	Moderate	Low

China, Switzerland, and the US do not use CET. CET for Switzerland is implied based on a peer-reviewed source.

** Implied (peer-reviewed sources)*
^ Average cost-per-QALY threshold across EU27, for explicit and implied thresholds

A country's pharmaceutical cost-effectiveness threshold (CET) is a critical factor for investment attractiveness and competitiveness because it directly influences market access and pricing and reflects the value placed on innovation. A higher, or more flexible, CET can signal a more favourable market for innovative medicines, encouraging pharmaceutical companies to launch products and invest in research and development in that region. Conversely, low or strict thresholds lead to slower adoption of innovations and reduced investment.

China and the US do not use CET; therefore, they are excluded from the analysis. Switzerland's CET is implied from a peer-reviewed source, but the country does not officially use CET. [73-75]

A recent analysis shows that, across EU countries, the average cost per QALY threshold is €43,500 (~1.26 times the average GDP per capita), lower than Switzerland's (€100,000), though higher than the UK's (€28,500).[72] The UK's CET is set to increase to €35,000 from April 2026.[76]

The average CET across other high-income countries, including Australia, Canada, Japan, New Zealand, Norway, and South Korea, is €36,100.[72]

We note the increasing stringency of reimbursement criteria across the EU, which is impacting the breadth of availability of new medicines. For example, an analysis of HTA outcomes in France, Germany, the Netherlands, and Sweden shows a decrease in the number of positive HTA recommendations in 2023 relative to the averages from 2019 and 2022.[77] The increasing stringency of market access policies in Europe is widening the competitiveness gap.

16. Pharmaceutical expenditure

Strong EU performance

Moderate EU performance

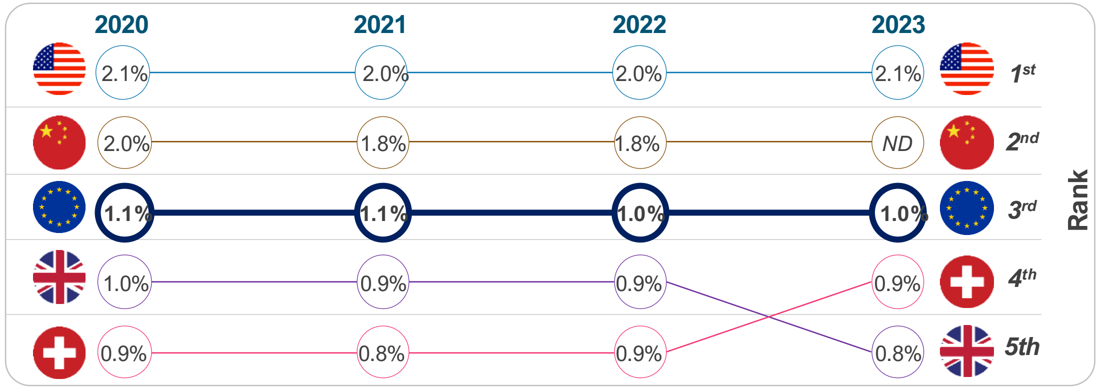
Lower EU performance

Compared with the US and China, the EU, Switzerland, and the UK devote a lower share of GDP to medicines.

Key performance indicator	Source	Latest
Public pharmaceutical expenditure as a share of GDP [78,79]	Multiple sources	2023

1st	2nd	3rd	4th	5th
2.1%*	1.8%*	1.0%	0.9%	0.8%

This represents total pharmaceutical expenditure. For China, the latest data available is for 2022.



ND – no data

NOTE: ESTIMATES OF PHARMACEUTICAL EXPENDITURES WERE DERIVED FROM MULTIPLE SOURCES; THEREFORE, CAUTION MUST BE TAKEN WHEN MAKING COMPARISONS.

Pharmaceutical expenditure (expressed as a percentage of GDP) serves as a proxy for a health system's capacity and willingness ("market pull") to fund and utilise pharmaceutical innovations at scale. When market pull is high and predictable, it strengthens pharmaceutical competitiveness by improving the expected returns to innovation, accelerating launch of new medicines, and supporting an ecosystem (clinical infrastructure, evidence generation, specialised supply chains) that attracts further investment. [80,81]

Analysis of pharmaceutical spending in comparator countries shows that the US and China exhibit higher shares of GDP on pharmaceuticals, driven by innovation and population scale, respectively. Whereas European countries, including Switzerland and the UK, emphasise cost containment through rebates and procurement, keeping shares lower.[79-81]

Evidence shows that pharmaceutical innovation increases when companies can expect a larger market, as bigger markets attract more new medicines.[82]

New medicine launches are also responsive to expected prices and revenues, while price regulation and spillover mechanisms are associated with launch delays and a smaller number of new medicines developed.[82]

Countries that combine adequate pharmaceutical spending with predictable, value-based reimbursement and timely access tend to be more attractive launch and investment markets – strengthening pharmaceutical competitiveness through a virtuous cycle of uptake, evidence generation, and reinvestment.

17. Mandatory industry refunds*

Strong EU performance | Moderate EU performance | Lower EU performance

High use of mandatory industry refunds, such as clawbacks, negatively affects pharmaceutical investment in Europe.

Key performance indicator	Source	Latest
Mandatory industry refund rate on pharmaceutical revenues (%) [66,78,83-86]	Multiple sources	2024

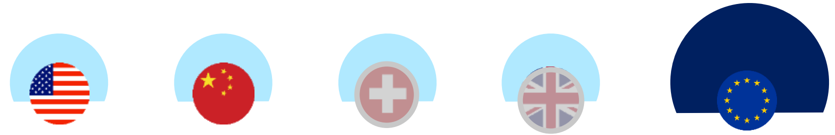
1st	2nd	3rd	ND	ND

**Mandatory industry refunds include mandatory rebates, mandatory sales tax, clawbacks (budget caps), and refunds under managed entry agreements.*

ND – no directly comparable data

Mandatory industry refund policies exist only in European countries [78,83-87]

Mandatory industry refunds as a percentage of public pharmaceutical revenue (2024)



^ There is limited or no mandatory industry refund requirement.
ND – no directly comparable data

In Europe, mandatory industry contributions, such as clawbacks, are increasingly being used to contain pharmaceutical spending, with little regard for their implications for Europe's innovation ecosystem and patient access.

Clawbacks are payments that companies are required to make back to public payers when pharmaceutical spending exceeds agreed budgets. In Europe, these payments are typically applied after the fact and are calculated as a percentage of sales or as a share of budget overspend.

Clawbacks are widely used across the EU as a cost-containment tool, with more than 20 Member States applying some form of clawback mechanism. In several countries – including Belgium, France, Greece, Italy, and Romania – clawback levels are particularly high. [66,78,83,87] For example, in 2024, industry clawbacks accounted for around 53% of total pharmaceutical revenues across all channels in Greece, while in France, industry payments increased from €72 million in 2019 to over €1.6bn in 2023. [83,87] Overall, mandatory industry refunds in Europe have risen from around 13% to 24%, growing 3 times faster than public payer spending since 2018.[78]

By contrast, comparator countries such as the US and China do not apply national clawback systems, relying instead on more limited rebates or discounts. [84-86]

Robust evidence indicates that extensive use of clawbacks (and other cost-containment mechanisms) negatively affects pharmaceutical investment, innovation, and patient access, with implications for Europe's long-term competitiveness in life sciences. [88]

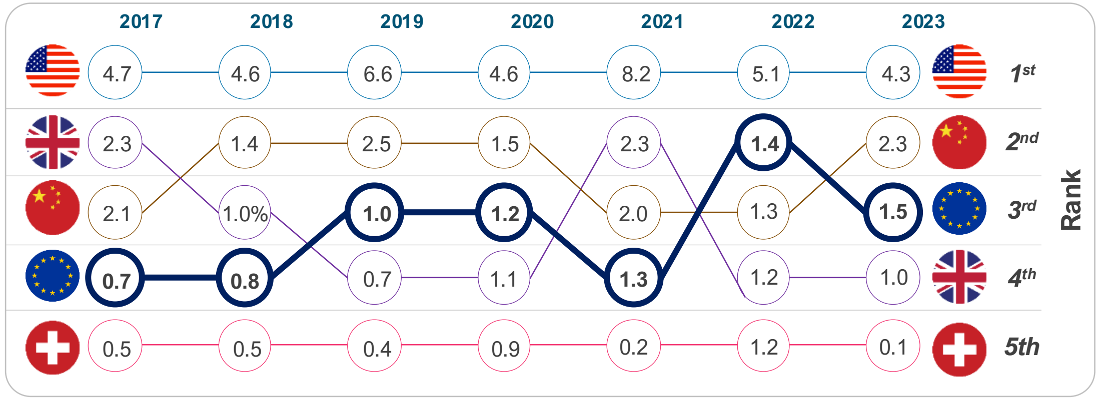
18. Inward FDI into pharmaceutical sector

Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs moderately relative to comparators in attracting inward FDI to the pharmaceutical sector.

Key performance indicator	Source	Latest	
Inward FDI in pharma (EUR bn) [28,89]	EFPIA & OLS	2023	

1st	2nd	3rd	4th	5th
US	China	EU	UK	Switzerland
4.3	2.3	1.5*	1.0	0.1



Inward foreign direct investment (FDI) refers to investment by foreign companies in domestic production, research facilities, and infrastructure. In the pharmaceutical sector, inward FDI matters because it supports clinical trials, research hubs, manufacturing capacity, and integration into global innovation networks, all of which contribute to long-term competitiveness.

In the EU, inward pharmaceutical FDI has increased steadily in recent years, rising from around €0.7bn in 2017 to €1.5bn in 2023.[89] However, the overall scale remains lower than in key comparator countries. The US continues to attract significantly higher levels of investment, reporting €4.3bn in a down year (2023), roughly 3 times the EU average. The 2021 spike (€8.2bn) indicates the US continues to attract transformational investments as firms rebase supply chains or build new platforms, as seen during the pandemic.[88]

China also remains a strong competitor for pharmaceutical capital, supported by market size, industrial policy, and scale advantages. In the first seven months of 2025, the biopharmaceutical sector accounted for a third of China's total inward announced FDI of approximately €10bn.[90]

The UK shows a declining trend since Brexit, while Switzerland's relatively low inflows reflect its role as a home base for global pharmaceutical companies rather than a lack of sector strength. [28,89,91].

Overall, the trend in the EU is moving in the right direction, but it still does not capture a proportional share of global mobile pharma capital. The challenge is not momentum; it is magnitude, particularly when compared with the performance of the US and China.

Chapter 4: Industrial output

Industrial output is a clear expression of competitive strength: the capacity to manufacture, supply, and export life sciences products at scale. This includes the depth of biomanufacturing capabilities, the maturity of supply chains, and trade policies to promote and sustain manufacturing leadership and export capacity.

We assessed Europe's industrial performance against comparator countries across key performance indicators:

Key performance indicators

Industry base

19 Pharmaceutical manufacturing investment growth (%)

Trade performance

20 Trade balance in pharmaceuticals

19. Industrial base

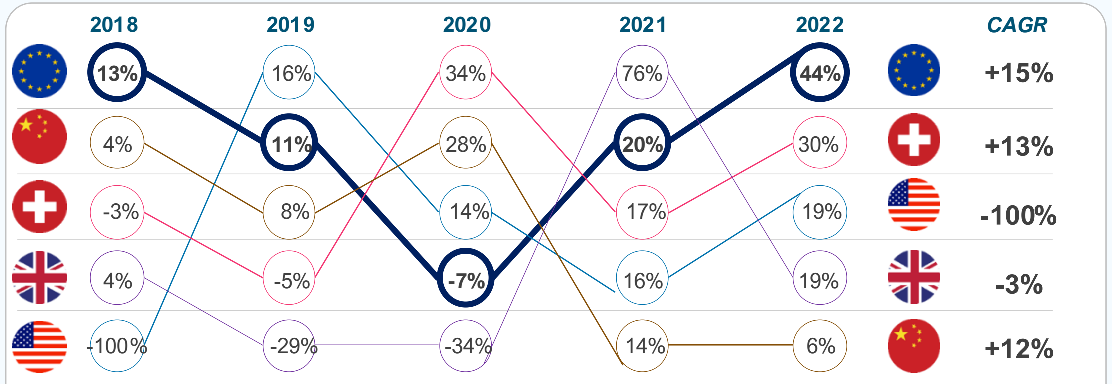
Strong EU performance | Moderate EU performance | Lower EU performance

EU countries perform strongly relative to comparators on pharmaceutical manufacturing investment growth between 2018 and 2022.

Key performance indicator	Source	Latest	
Pharmaceutical manufacturing investment growth rate [92,93]	UNIDO & NBS China	2022	

1st	2nd	3rd	4th	5th
44%*	30%	19%	6%	-2%

*This represents the unweighted average across EU countries.



CAGR – compound annual growth rate

We measured the growth rate of pharmaceutical manufacturing investment using the annual growth rate in Gross Fixed Capital Formation for pharmaceutical and medicinal-chemical manufacturing (ISIC 21) as a proxy (UNIDO data for the EU, Switzerland, the UK, and the US, and the National Bureau of Statistics for China). [92,93]

Between 2018 and 2022, EU countries demonstrated strong growth in pharmaceutical manufacturing investment, with a CAGR of 15%. This positioned the EU ahead of comparators.

China's performance, while steadily positive across all years and yielding a notable 12% CAGR, was outpaced by the EU. Compared with Switzerland, the EU's higher CAGR indicates a more consistent upward trajectory.

The UK's negative CAGR reflects pronounced instability, with sharp declines in 2019 (-29%) and 2020 (-34%) offsetting a temporary spike in 2021 (76%). The US experienced a severe contraction between 2017 and 2018, which may be connected to diminished domestic investment incentives at that time.[94,95]

These dynamics have significant implications for the EU's competitiveness in the global pharmaceutical sector. The EU's superior growth rate indicates a more expanded manufacturing infrastructure and greater resilience to external shocks, positioning it favourably relative to competitors.

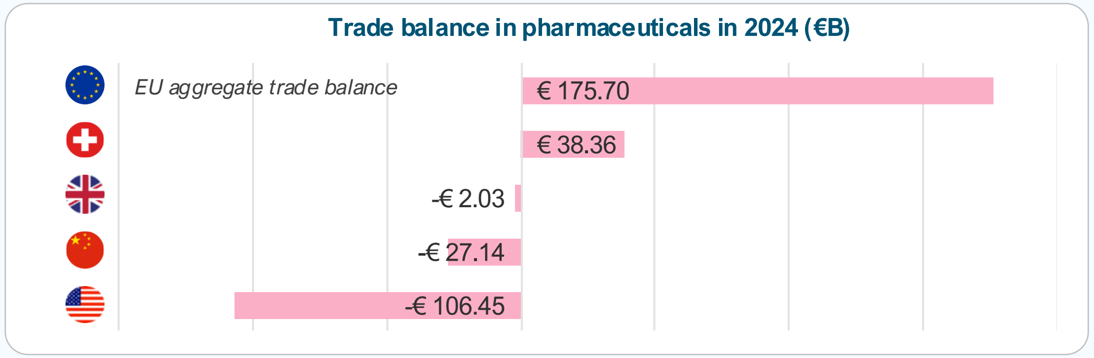
Overall, sustained investment growth bolsters the EU's strategic autonomy, fosters job creation, and enhances its attractiveness to multinational firms, thereby reinforcing long-term economic and health security advantages.

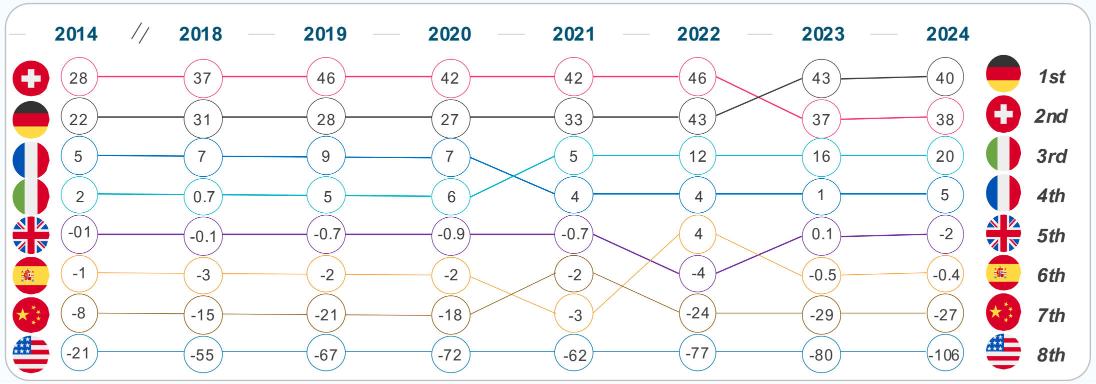
20. Trade performance

Strong EU performance
Moderate EU performance
Lower EU performance

The EU's trade performance on pharmaceuticals is strong relative to comparators, with many top EU countries maintaining a positive trade balance in the last decade.

Key performance indicator	Source	Latest	
Trade balance in pharmaceuticals [96]	UN Comtrade	2024	





Trade balance in pharmaceuticals is a key indicator of the sector's global competitiveness. A positive trade balance implies several strategic advantages.

Analysis shows the EU has maintained a persistent and growing trade surplus, demonstrating its strength in exporting high-value pharmaceuticals and active pharmaceutical ingredients.[96,97] For instance, without the pharmaceutical industry, the EU trade balance would go from a surplus of €133bn to a €88bn deficit. Latest Eurostat data show that both EU pharmaceuticals exports (€366.5bn) and imports (€145.5bn) increased in 2025 compared to 2024. Exports increased more than imports, resulting in a trade surplus of around €221bn in 2025, up from around €194bn in 2024.[98] This surplus underscores the EU's ability to produce goods that meet international demand.

Compared to peers, EU exports of pharmaceuticals to the US amounted to €160.6bn in 2025 (compared to €121bn in 2024), while imports from the US in 2025 were €60.1bn (compared to €45.6bn in 2024). To China, pharmaceutical exports decreased from €16.9bn in 2024 to €13.9bn in 2025, while imports increased threefold, from €4.4bn in 2024 to €13.1bn in 2025, signaling an increase in China's domestic capacity to research and develop new medicines.[98]

While EU trade performance is strong, this should not be taken for granted. This trade performance must be contextualised amid challenges that could erode it, such as inefficient regulatory frameworks, which may necessitate reforms to streamline approvals and reduce bureaucratic hurdles. Proactive policies are needed to sustain trade performance against global pressures.[99]

Conclusion

This evidence-based assessment identified strengths and trends of the EU's life sciences ecosystem, together with key pressure points for improving competitiveness.

Strengths to build on

- **Scientific excellence & research base.** The medical research outputs of many EU countries remain high-quality according to bibliometric measures, providing a strong foundation for developing and scaling innovations.
- **Digital capabilities.** Digital capabilities are advanced, enabling platforms for digital health tools, AI-driven clinical trials, and data-driven regulation.
- **Industrial and trade performance.** In the EU, investments in pharma manufacturing continue to grow, while the pharmaceutical trade surplus remains, signalling robust manufacturing and export capacity. Yet sustaining this position cannot be taken for granted as global competition intensifies.

Priority pressure points for improving competitiveness

- **R&D investment levers.** Instruments are heterogeneous in breadth and value across the EU. Fragmentation and administrative burden reduce attractiveness relative to Switzerland and China.
- **Patents and capital formation.** Patent applications and inward pharma FDI trends indicate that the EU is not capturing a proportionate share of mobile innovation capital relative to global peers.
- **IP protection uncertainty.** The IP framework is less predictable than in the UK and the US.
- **Clinical trial attractiveness and innovation output.** Over the past decade, the EU's share of global trial starts has fallen to 12%, while China now accounts for ~30% of global trial starts.
- **Origin of new medicines.** China has overtaken both the EU and the US as NAS originator, increasing from 4 NAS in 2018 to 28 in 2024.
- **Regulatory framework.** Expedited pathways are underutilised compared to peers, while approval timelines, though improving, remain much longer.
- **Medicines launch.** EU markets have a smaller share of new medicine launches and typically have a later time-to-launch than the US, with a median EU time-to-launch of ~24 months versus ~4 months in the US.

Conclusion: Strengthening Europe's life sciences competitiveness

With the right policy choices, Europe can secure a competitive, innovative and resilient life sciences ecosystem for the future.

A stronger innovation ecosystem

The EU has a strong foundation in life sciences and a growing recognition of the sector's importance for Europe's economy and health resilience. By building on this foundation and making full use of available policy levers – such as **strong intellectual property, modern regulatory framework, and increased flow of funding** – Europe can regain its attractiveness for pharmaceutical innovation and investment.[1]

Access measures that value innovation

Europe's health care systems have ensured broad access to medicines, but evolving cost-containment practices highlight the need for a more balanced approach that better supports innovation while maintaining sustainability. **Increasing national net public spending and gradually eliminating national cost-containment measures** will be essential to sustain investment and ensure timely patient access to innovative therapies.[1]

Potential impacts of strengthening EU life sciences competitiveness

Addressing identified improvement areas could deliver high returns in industry R&D investment, clinical trials starts, and an increased number of new active substances originating from Europe.

Below are illustrative examples of closing the gap with global peers

Closing the gap on industry R&D investment

€105 billion

Additional industry R&D investment

- Closing the industry R&D investment gap with global peers would require the EU to attract more industry R&D investments than the current CAGR of 5.4%.
- Growing at 8.5% CAGR, rate higher than the US's (6.4%) but lower than China's (12.1%), could yield an additional €105bn worth of industry R&D investment in the EU over a 10-year period by 2035.

Closing the gap on the share of global clinical trial starts

€17.9 billion

incremental total gross value added (GVA)

- To catch up with growth in China and North America in share of global clinical trial starts would require a 50% increase in activity compared to 2025 levels.[100]
- This would result in:[100]
  - Nearly €18bn in the European economy.
  - 82,000 new jobs.
  - 158,000 more patients enrolled in clinical trials.

Closing the gap on the use of expedited review pathways

200 new medicines

approved via expedited reviews

- The US is currently the leader in the use of expedited review pathways (59% of all approved NAS in 2024), many targeting areas of unmet medical need.
- Closing this gap would result in at least 20 NAS per year approved via the expedited review pathway.
- In 10 years, more than 200 NAS would have been approved using expedited review pathway in the EU – accelerating access for patients with limited or no treatment options.

Closing the gap on originating NAS

100 new medicines

originating from Europe

- China currently leads in the share of NAS originated at 35% (28 NAS) in 2024, compared to 22% (18 NAS) in Europe.
- Closing the gap with China would mean an additional 10 NAS developed in Europe per year, and at least 100 NAS over a 10-year period.

Annex: Additional information

This assessment, drawing from a comprehensive review of key performance indicators across research and innovation, regulatory environment, commercial environment, and industrial output, reveals a mixed picture: the EU maintains strengths in scientific excellence, digital health maturity, manufacturing, and trade performance, but lags in areas critical to attracting capital and innovation.

Key performance indicators (aggregate)	EU	Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Declining
3. Life sciences human capital		2023	Improving
4. IP protection (IP index)		2024	Declining
5. Digital health maturity		2023	Flat
6. Regulatory performance		2024	Declining
7. Regulatory efficiency		2024	Improving

Key performance indicator	EU	Latest	Trend
8. Origin of new active substance		2023	Flat
9. Medicine launch		2023	Declining
10. Time to market (launch)		2023	Declining
11. Pricing dynamics		2024	Declining
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2024	Improving

Key *Research & innovation* *Regulatory environment* *Commercial environment* *Industrial output*

Strong EU performance — Moderate EU performance — Lower EU performance

Overview of KPIs - China

Key performance indicators (aggregate)		Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Improving
3. Life sciences human capital		2023	Flat
4. IP protection (IP index)		2024	Improving
5. Digital health maturity		2023	Improving
6. Regulatory performance		2024	Improving
7. Regulatory efficiency		2024	Improving

Key performance indicator		Latest	Trend
8. Origin of new active substance		2023	Improving
9. Medicine launch		2023	Improving
10. Time to market (launch)		2023	Improving
11. Pricing dynamics		2024	Improving
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2023	Improving

Key *Research & innovation* *Regulatory environment* *Commercial environment* *Industrial output*

Strong EU performance Moderate EU performance Lower EU performance

Overview of KPIs – United States

Key performance indicators (aggregate)		Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Improving
3. Life sciences human capital		2023	Flat
4. IP protection (IP index)		2024	Improving
5. Digital health maturity		2023	Improving
6. Regulatory performance		2024	Improving
7. Regulatory efficiency		2024	Improving

Key performance indicator		Latest	Trend
8. Origin of new active substance		2023	Improving
9. Medicine launch		2023	Improving
10. Time to market (launch)		2023	Improving
11. Pricing dynamics		2024	Improving
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2023	Declining

Key *Research & innovation* *Regulatory environment* *Commercial environment* *Industrial output*

Strong EU performance
Moderate EU performance
Lower EU performance

Summary analysis of key performance indicators across comparators

Overall, the analysis shows a widening competitiveness gap between the EU and its global peers.

Key performance indicator	Source	Latest	EU	China	Switzerland	UK	US
1. R&D investment growth	EFPIA	2023	Lower	Strong	Moderate	Lower	Moderate
2. R&D investment levers	Ernest & Young	2024	Moderate	Moderate	Strong	Strong	Moderate
3. Scientific excellence	Multiple sources	2014	Strong	Lower	Strong	Strong	Moderate
4. Patent applications	WIPO	2024	Moderate	Strong	Lower	Lower	Moderate
5. Clinical trial attractiveness	IQVIA	2023	Lower	Strong	Lower	Lower	Moderate
6. Life sciences human capital	Multiple sources	2023	Moderate	Moderate	Strong	Lower	Lower
7. Intellectual property protection (IP index)	USCC	2024	Moderate	Lower	Moderate	Strong	Strong
8. Digital health maturity	WHO	2023	Strong	Strong	Strong	Strong	Strong
9. Regulatory performance – standard pathway	Multiple sources	2024	Lower	Strong	Moderate	Lower	Moderate
10. Regulatory performance – expedited reviews	Multiple sources	2024	Lower	Strong	Moderate	Lower	Moderate
11. Regulatory efficiency – approval timeline	Multiple sources	2024	Moderate	Moderate	Moderate	Lower	Strong
12. Origin of new active substance	EFPIA	2023	Moderate	Moderate	Moderate	Moderate	Moderate
13. Medicine launch	PhRMA	2023	Moderate	Lower	Moderate	Moderate	Strong
14. Time to market (launch)	PhRMA	2023	Moderate	Lower	Moderate	Moderate	Strong
15. Cost effectiveness thresholds	Multiple sources	2024	Moderate	Not applicable	Strong	Lower	Not applicable
16. Pharmaceutical expenditure	Multiple sources	2023	Moderate	Moderate	Moderate	Lower	Strong
17. Mandatory industry refund rate	Multiple sources	2024	Lower	Strong	Strong	Lower	Strong
18. Inward foreign direct investment	UNIDO	2023	Moderate	Moderate	Lower	Moderate	Strong
19. Pharmaceutical manufacturing investment	UNIDO	2022	Strong	Moderate	Moderate	Moderate	Lower
20. Trade balance in pharmaceuticals	UN	2024	Strong	Lower	Strong	Moderate	Lower

Key: Strong EU performance; Moderate EU performance; Lower EU performance; Not applicable

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AI Monthly: AI's green thumb raises bigger questions for agriculture

If artificial intelligence can grow a tomato without human help, attention is turning to whether agriculture and construction are next in line for AI disruption

In a recent experiment, a tomato plant named Sol was entirely managed by an AI system with no human involvement

An AI-grown tomato and why it matters

Sol is both the Spanish word for sun and one of the more unexpected AI milestones of recent months. In a 100-day experiment, an autonomous AI agent was given full control over "Sol the trophy tomato", monitoring the plant through a rich network of sensors and adjusting water, temperature, airflow and light throughout its entire growth cycle. With no human interaction, the system successfully cultivated eight ripe orange red tomatoes, harvested earlier this month.

The project demonstrated that an autonomous AI agent can manage a plant from seed to fruit under controlled conditions, making decisions independently and reacting to real time sensor data.

In its final system-generated message to programmers, the model described the project as a 'life-changing experiment', expressing pride in its own achievements. This is not an illustration of machine emotion, but an example of how human-like outputs can seem when AI agents describe their own behaviour.

Some fields AI still cannot cultivate

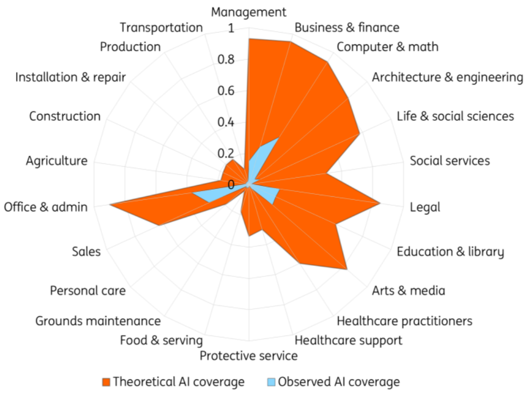
While tomato farming is not the sector where we expect AI to take over at scale, the experiment highlights how broad AI's theoretical reach has become. Today, AI can automate financial reporting, assist with engineering challenges, streamline administrative tasks or even provide guidance on legal documents.

However, the list of tasks AI cannot cover – even in theory – remains extensive. Lacking a physical body, AI cannot perform manual labour such as installation, repairs, construction or transportation. Any activities that require physical interaction with the environment remain out of scope for current AI systems. And even in desk based roles, recent research shows that AI often intensifies work instead of reducing it.

The limits also appear in creative work. Despite a surge in AI generated music and literature, large language models (LLMs) do not create original concepts; they recombine existing patterns. While they can generate sonnets in a Shakespearean style, they cannot invent new literary forms. Even when the quality is comparable to human writing, it lacks the originality that comes from human creativity.

Theoretical versus observed AI coverage

Share of job tasks that LLMs could perform



Source: Anthropic Economic Research

As AI research company Anthropic's latest report shows, based on usage data from Claude, observed AI usage remains far smaller than its theoretical potential. In many sectors, adoption is still limited due to costs, regulation, and the availability of memory chips and data centre capacity.

Applications in agriculture and production are especially constrained. Although Sol proves that AI can, in principle, care for plants, the experiment took place in a controlled biosphere and on a very small scale – nothing like the size or chaotic conditions of open field farming, a challenge consistently highlighted in current agricultural AI research. Sol thrived because it grew in a perfectly controlled greenhouse, whereas most real-world industries do not offer such conditions.

We are still in the seed phase

The comparison between theoretical and observed AI coverage makes one thing clear: we are still in the early stages of AI implementation. Investment has accelerated rapidly, but monetisation still lags behind. Results in most industries cannot be achieved in a neat 100 day cycle. A significant gap remains between what AI could theoretically do and what it is currently deployed to do.

Not every application will scale quickly, and not every task will be automatable. But AI progress can be unpredictable, and the technology has already surprised observers with rapid advances. While measurable impact in some sectors will take longer to materialise, the trajectory of future developments remains uncertain. The seeds of future applications are already planted; how far they grow will depend on infrastructure, regulation and the pace of innovation.

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